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(12) **United States Patent**
Whetsel

(10) **Patent No.:** **US 12,416,670 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **FALLING CLOCK EDGE JTAG BUS ROUTERS**

G01R 31/318552 (2013.01); **G01R 31/318555** (2013.01); **G01R 31/318558** (2013.01)

(71) Applicant: **TEXAS INSTRUMENTS INCORPORATED**, Dallas, TX (US)

(58) **Field of Classification Search**
None
See application file for complete search history.

(72) Inventor: **Lee D. Whetsel**, Parker, TX (US)

(56) **References Cited**

(73) Assignee: **TEXAS INSTRUMENTS INCORPORATED**, Dallas, TX (US)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(Continued)

(21) Appl. No.: **18/211,369**

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(22) Filed: **Jun. 19, 2023**

“Programming a Flash-Based MSP430 Using the JTAG Interface” by Markus Koesler, Franz Graf, Zack Albus, Sep. 2002.

(65) **Prior Publication Data**

US 2023/0333163 A1 Oct. 19, 2023

Primary Examiner — Fahmida Rahman

Related U.S. Application Data

(74) *Attorney, Agent, or Firm* — Carl G. Peterson; Frank D. Cimino

(62) Division of application No. 17/158,438, filed on Jan. 26, 2021, now Pat. No. 11,680,985, which is a division of application No. 16/776,667, filed on Jan. 30, 2020, now Pat. No. 10,935,601, which is a division of application No. 16/228,067, filed on Dec. 20, 2018, now Pat. No. 10,585,144, which is a division of application No. 15/499,362, filed on Apr. 27, 2017, now Pat. No. 10,197,628, which is a division of application No. 15/179,117, filed on Jun. 10, 2016, now Pat. No. 9,671,463, which is a division
(Continued)

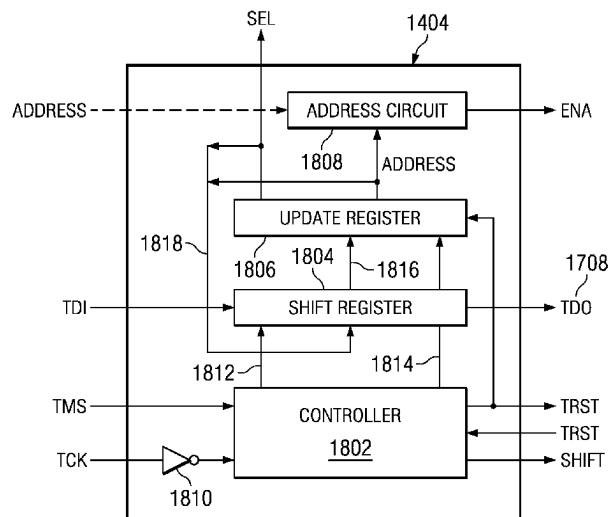
(57) **ABSTRACT**

A falling edge controller includes a controller having an inverted TCK (Test Clock) input, a TMS (Test Mode Select) input, a shift register control output, an update register control output, and a shift output; a shift register having a TDI (Test Data In) input, a shift register control input coupled to the shift register control output, address inputs, a select input, address and select outputs, and a TDO (Test Data Out) output; an update register having address and select inputs coupled to the address and select outputs, an update register control input coupled to the update register control output, address outputs coupled to the address inputs, and a select output coupled to the select input; and address circuitry having address inputs coupled to the address outputs, and having an enable output.

(51) **Int. Cl.**
G01R 31/317 (2006.01)
G01R 31/3177 (2006.01)
G01R 31/3185 (2006.01)

(52) **U.S. Cl.**
CPC **G01R 31/318597** (2013.01); **G01R 31/31723** (2013.01); **G01R 31/3177** (2013.01);

20 Claims, 28 Drawing Sheets



Related U.S. Application Data

of application No. 14/054,039, filed on Oct. 15, 2013, now Pat. No. 9,395,412, which is a division of application No. 12/968,966, filed on Dec. 15, 2010, now Pat. No. 8,589,714.

- (60) Provisional application No. 61/288,055, filed on Dec. 18, 2009.

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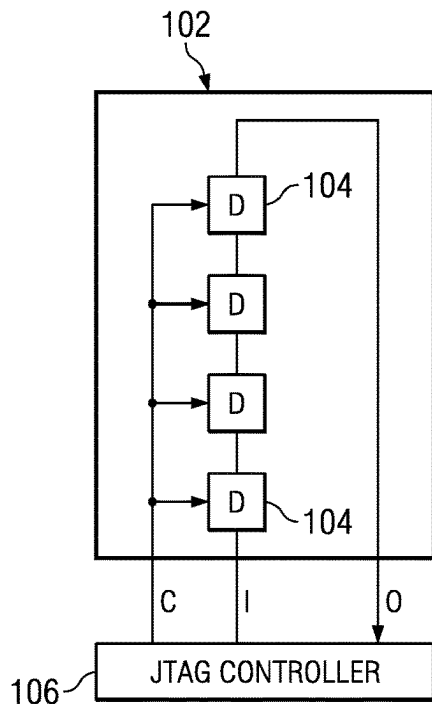


FIG. 1
(PRIOR ART)

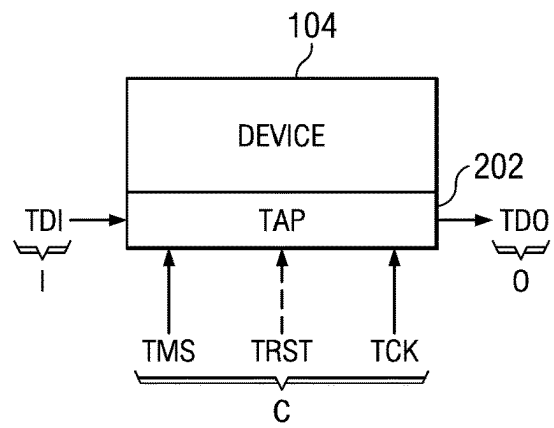


FIG. 2
(PRIOR ART)

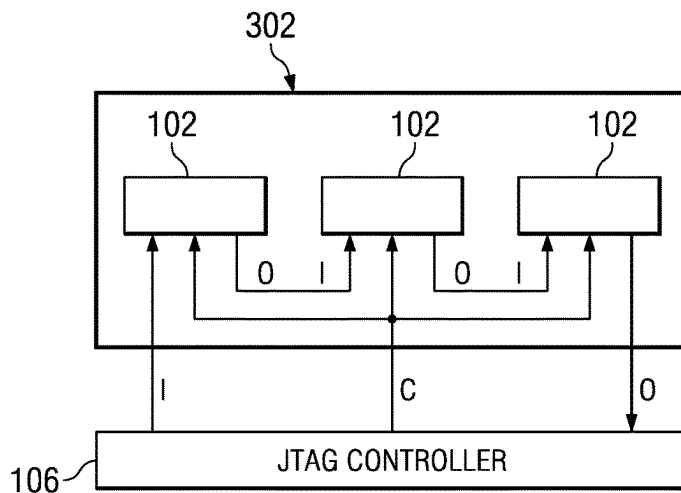


FIG. 3
(PRIOR ART)

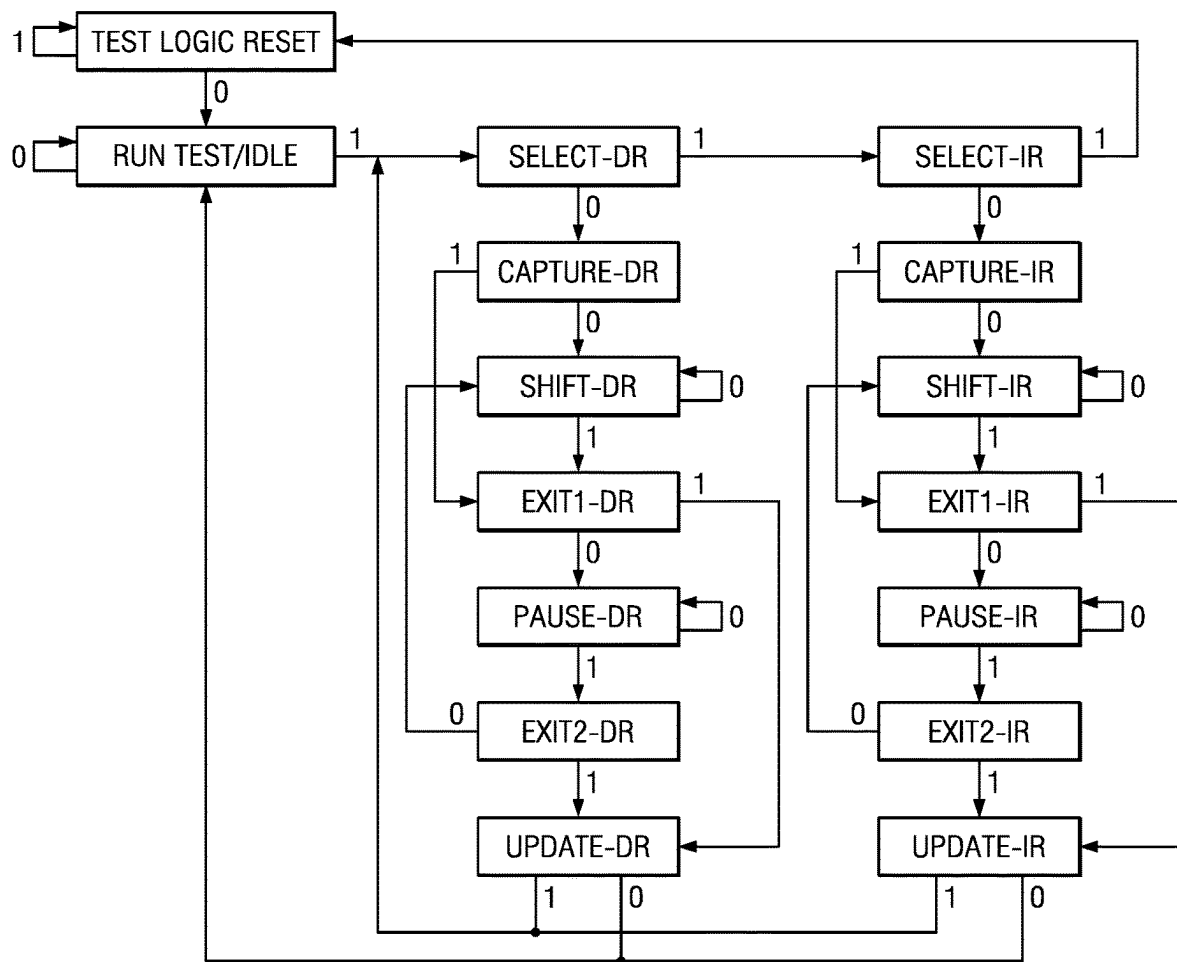


FIG. 4
(PRIOR ART)

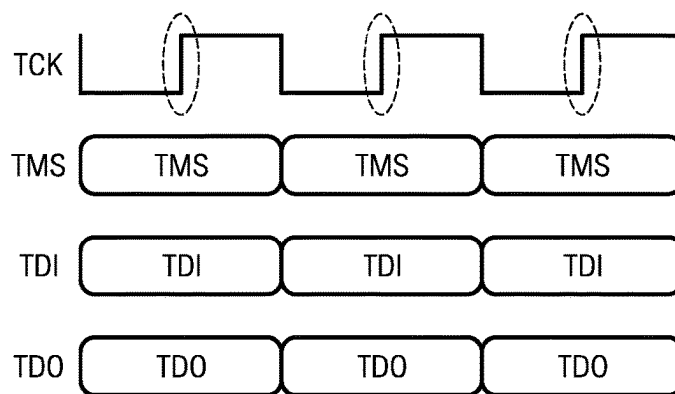


FIG. 5
(PRIOR ART)

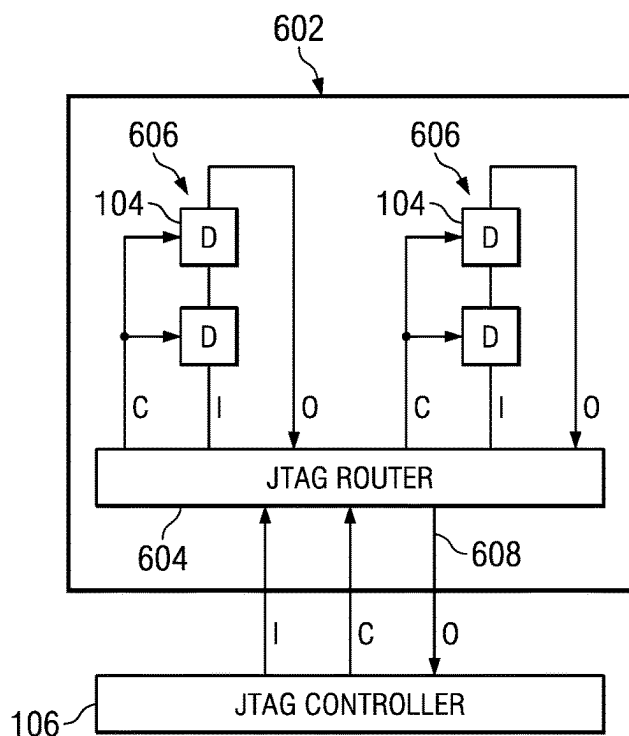


FIG. 6
(PRIOR ART)

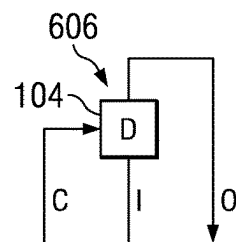


FIG. 6A

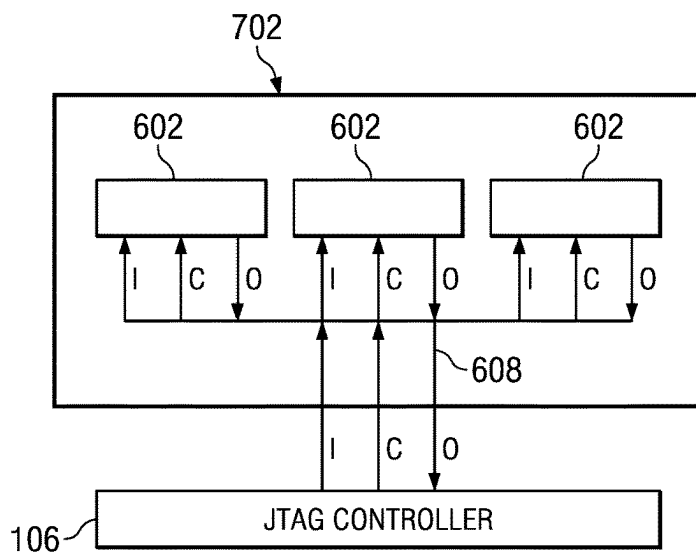


FIG. 7
(PRIOR ART)

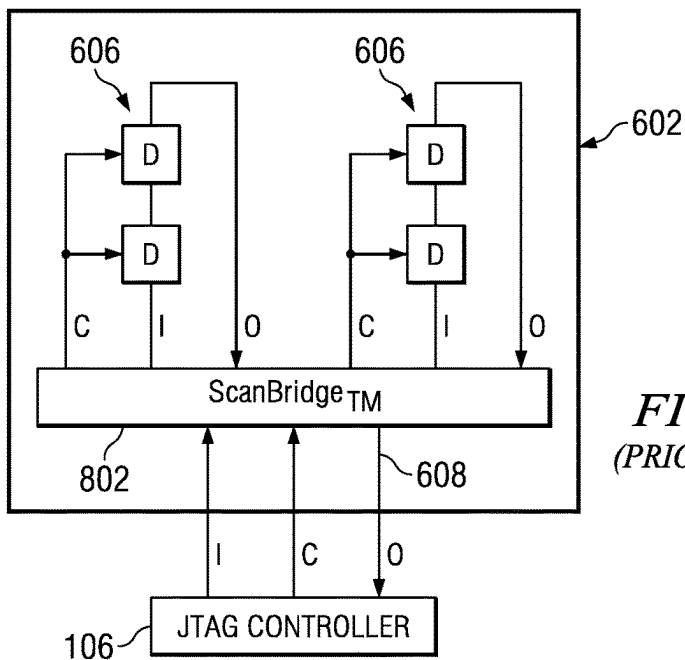


FIG. 8
(PRIOR ART)

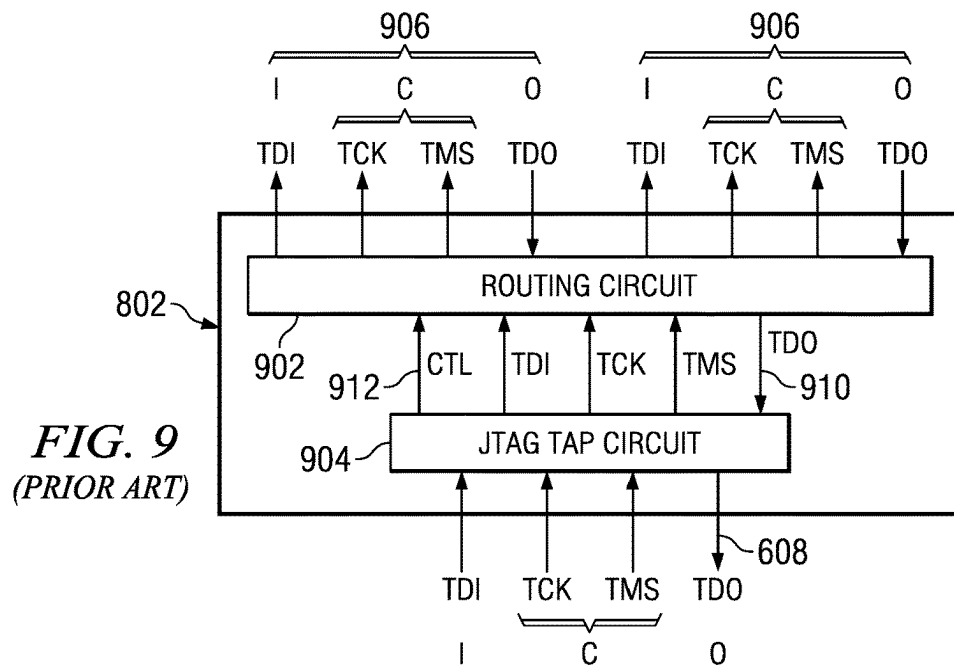


FIG. 9
(PRIOR ART)

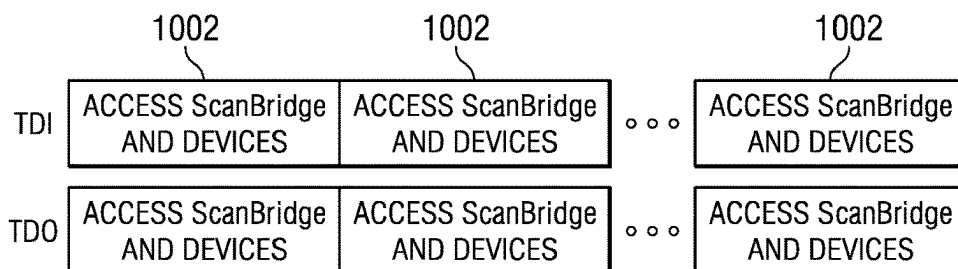


FIG. 10
(PRIOR ART)

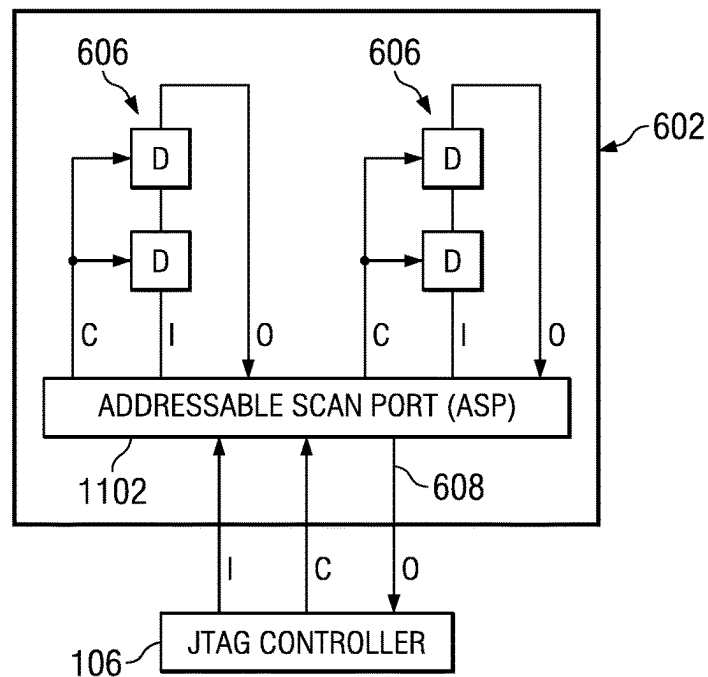


FIG. 11
(PRIOR ART)

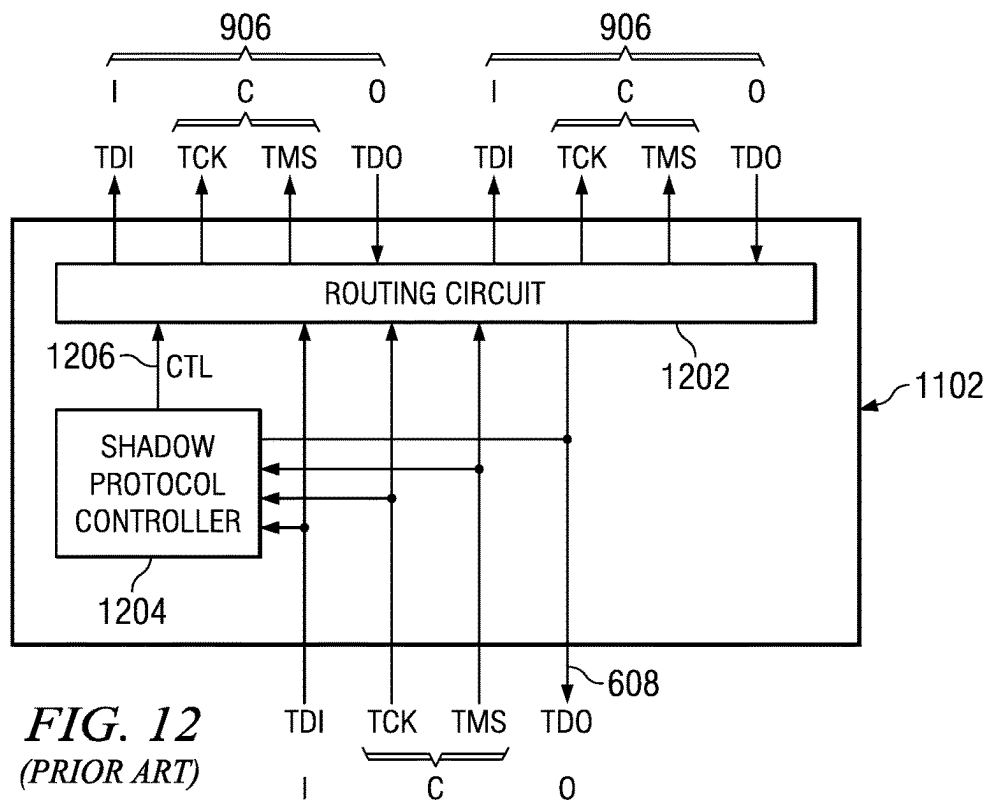


FIG. 12
(PRIOR ART)

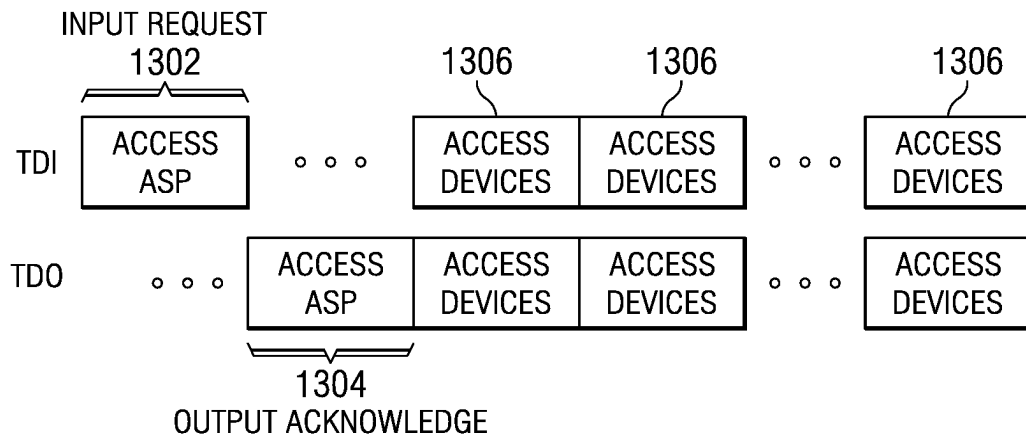


FIG. 13
(PRIOR ART)

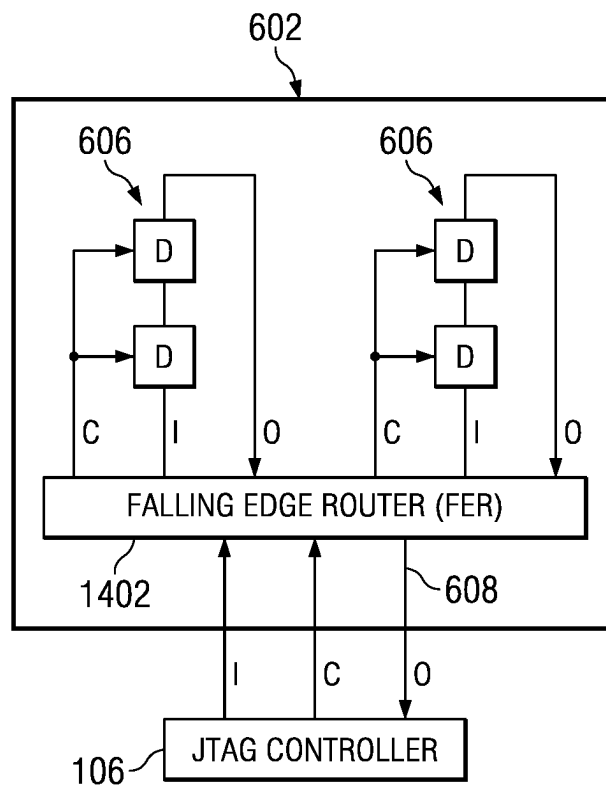
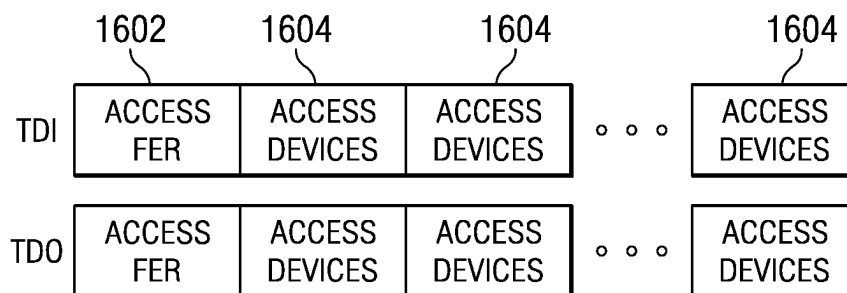
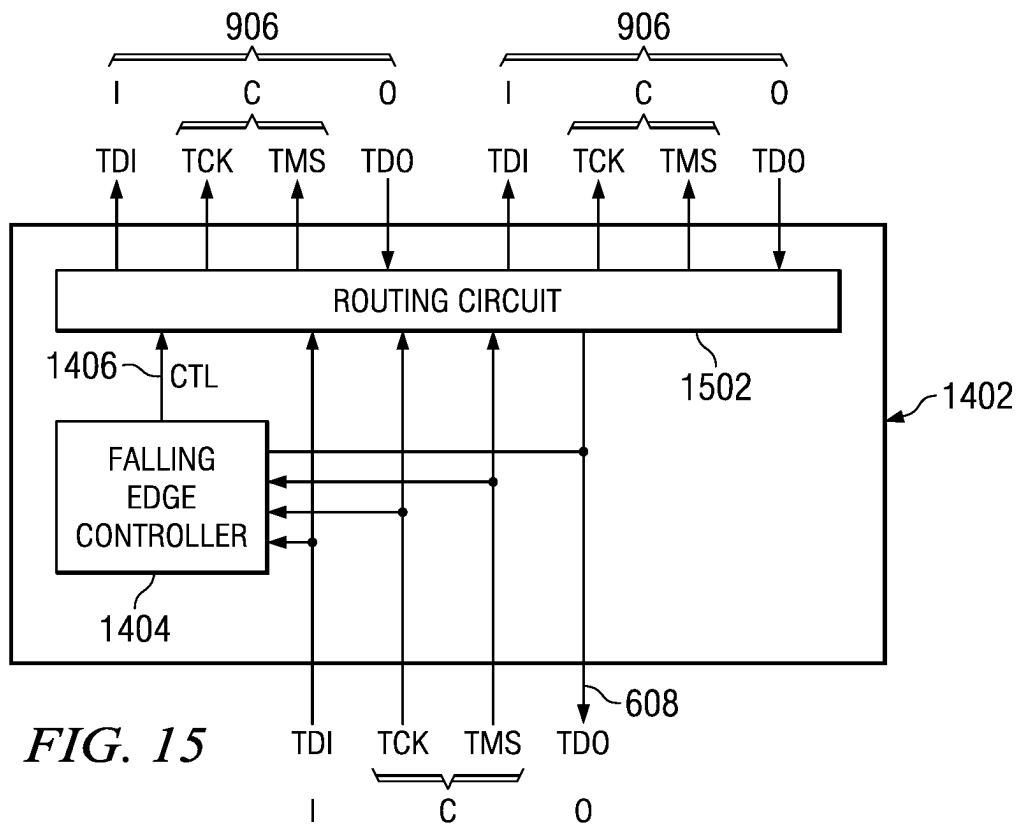
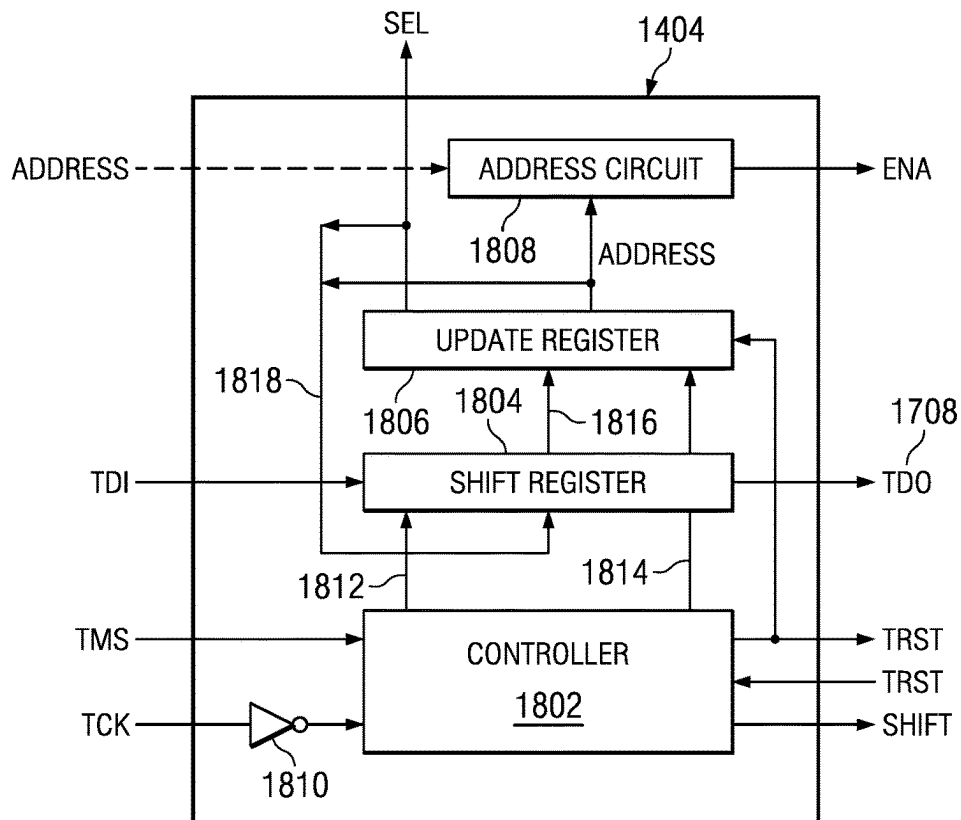
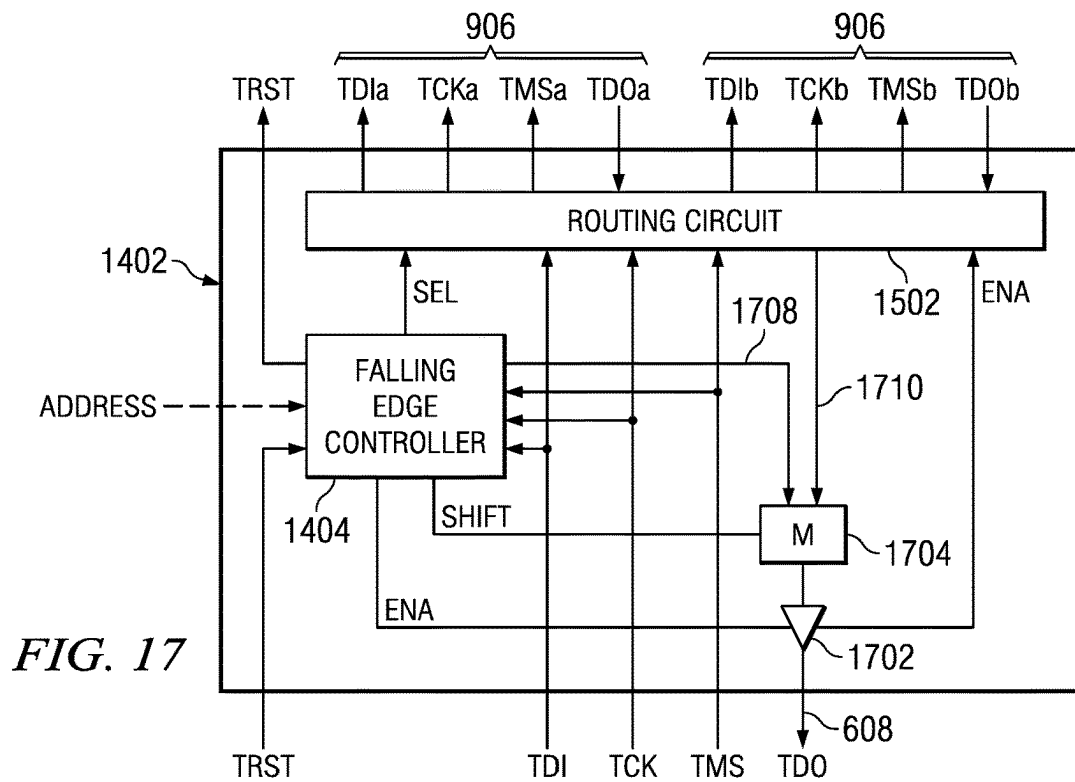


FIG. 14





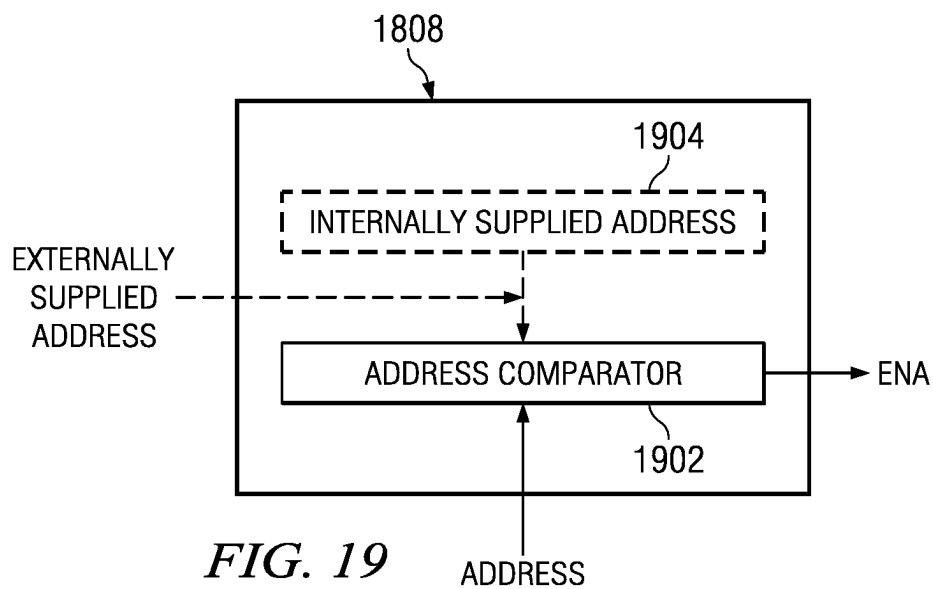


FIG. 19

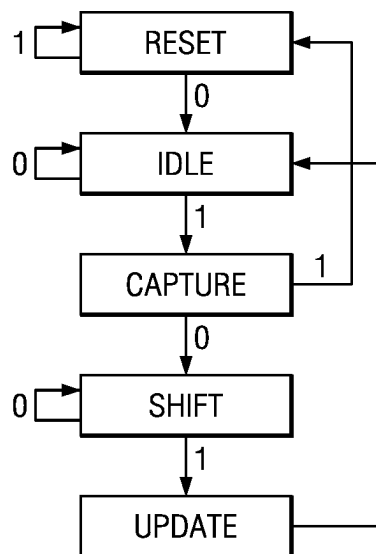


FIG. 20

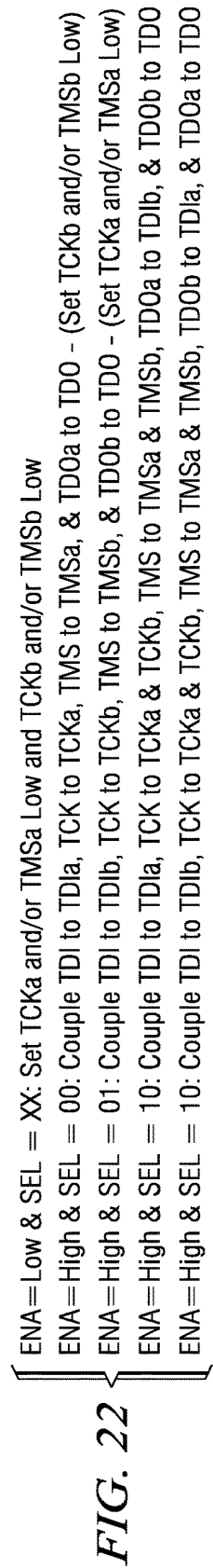
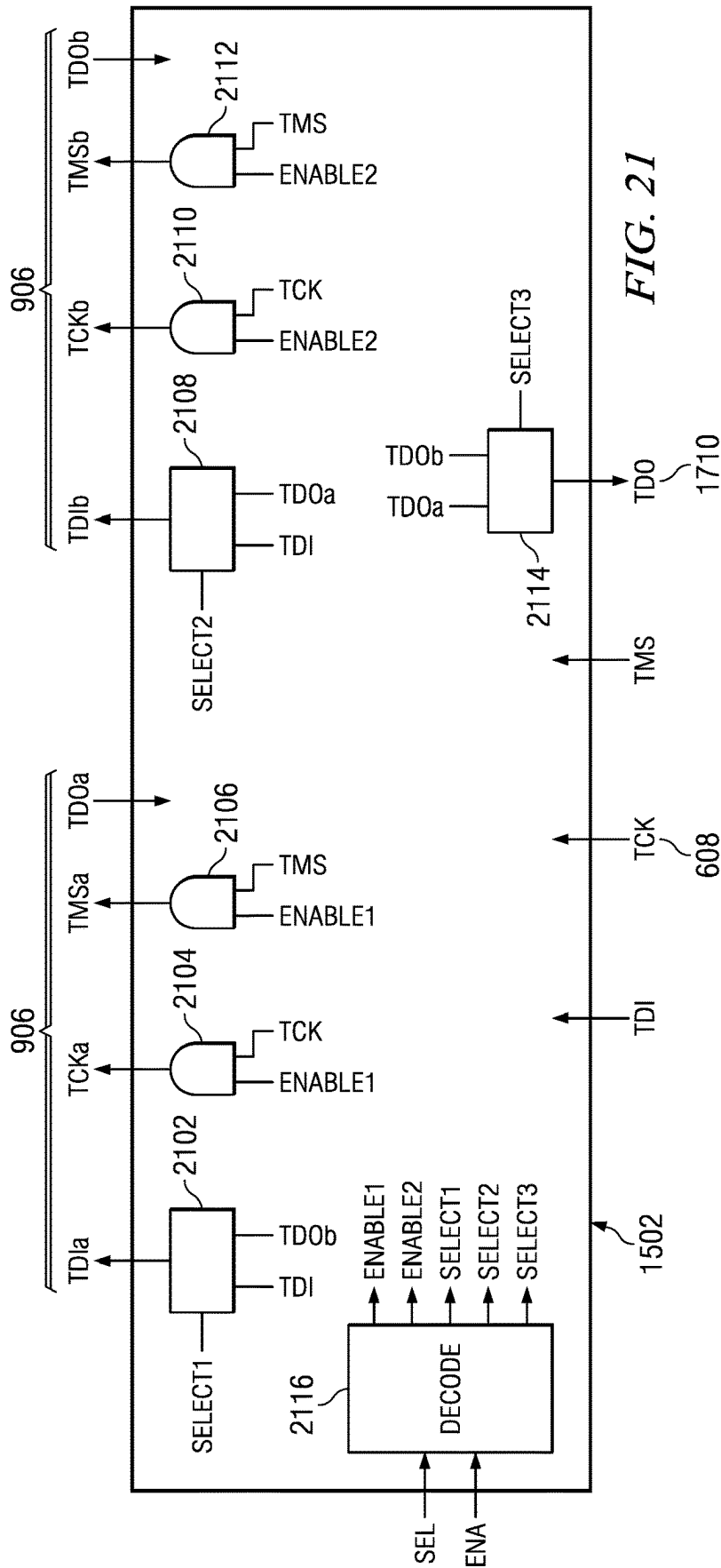
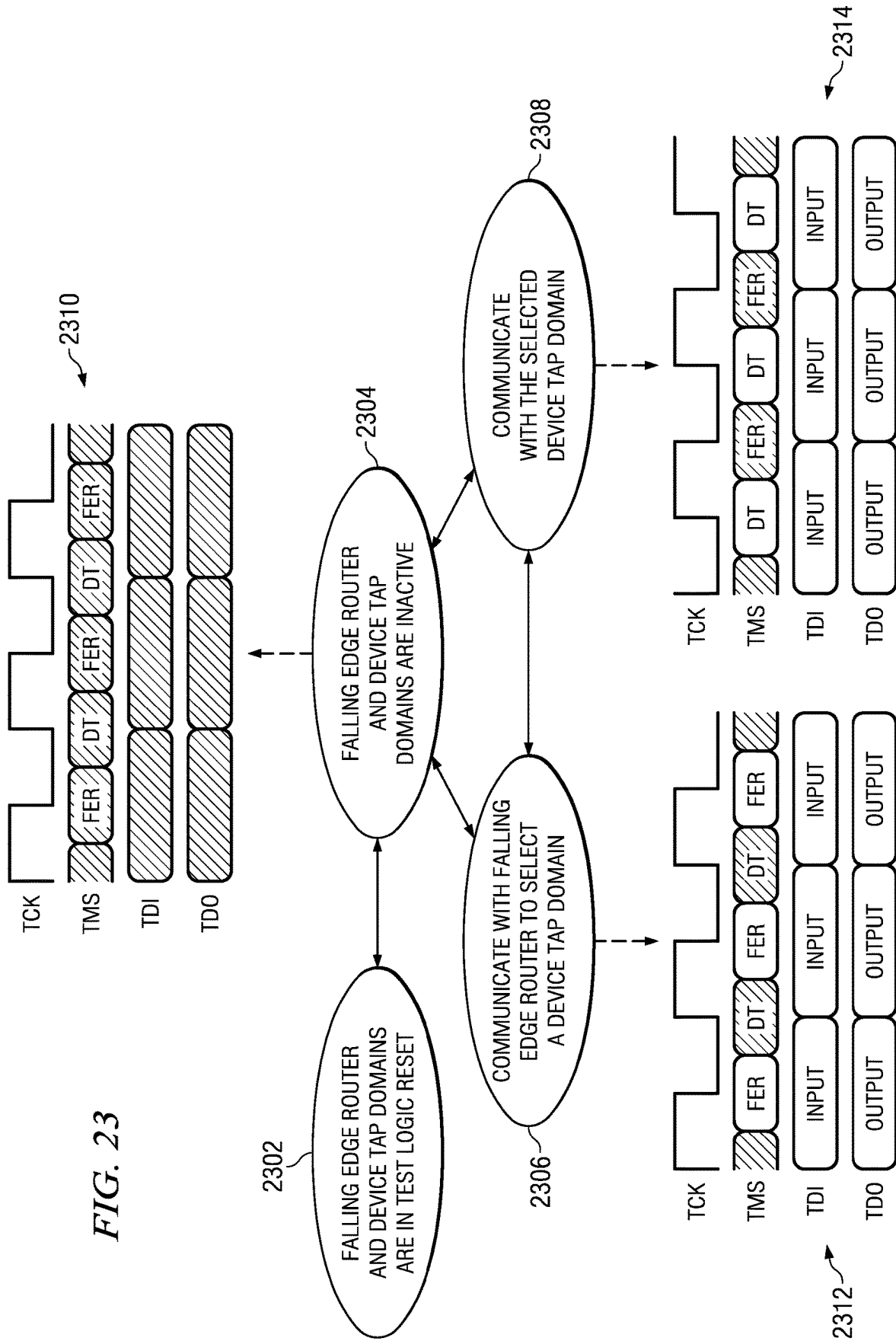


FIG. 23



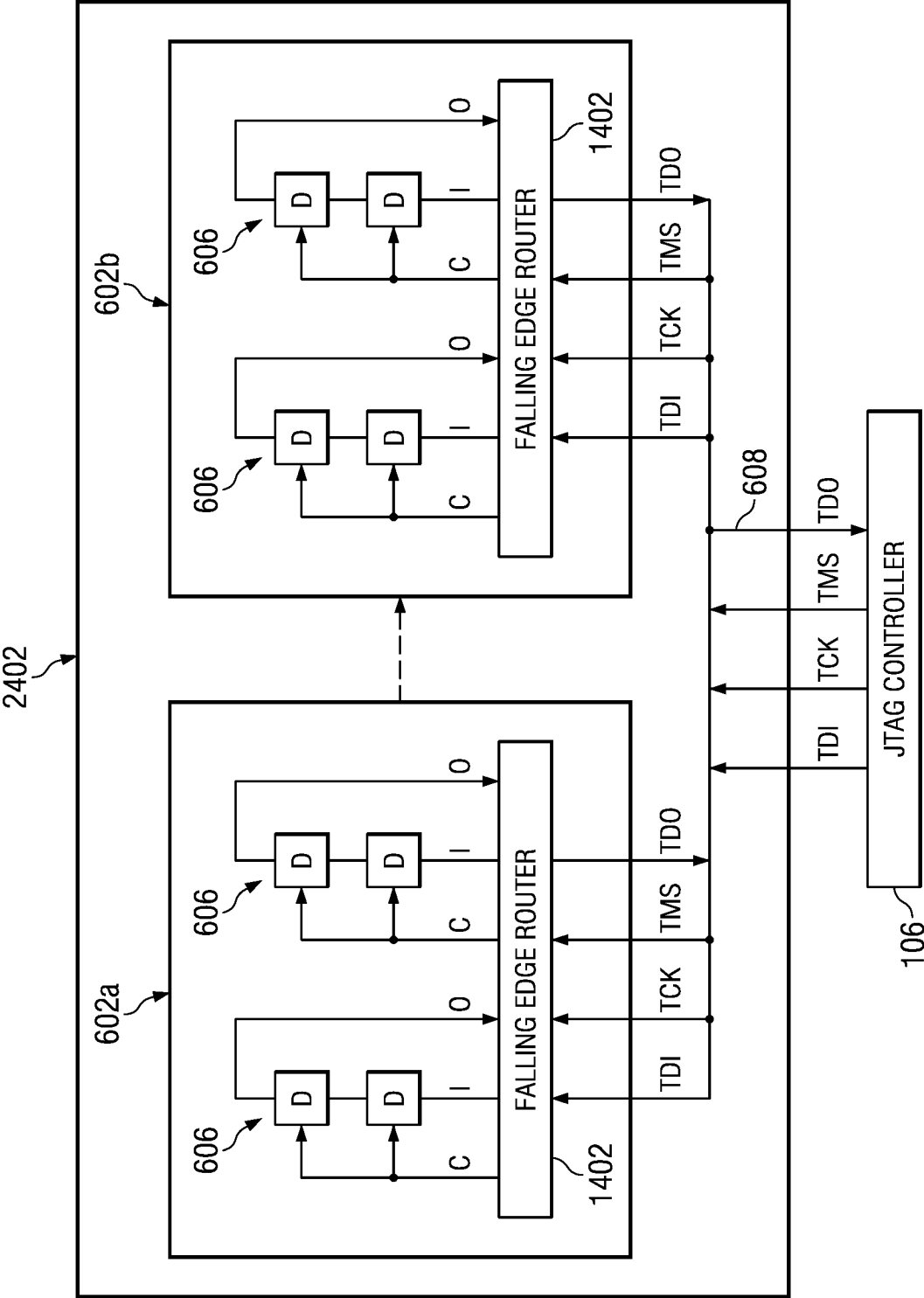


FIG. 24

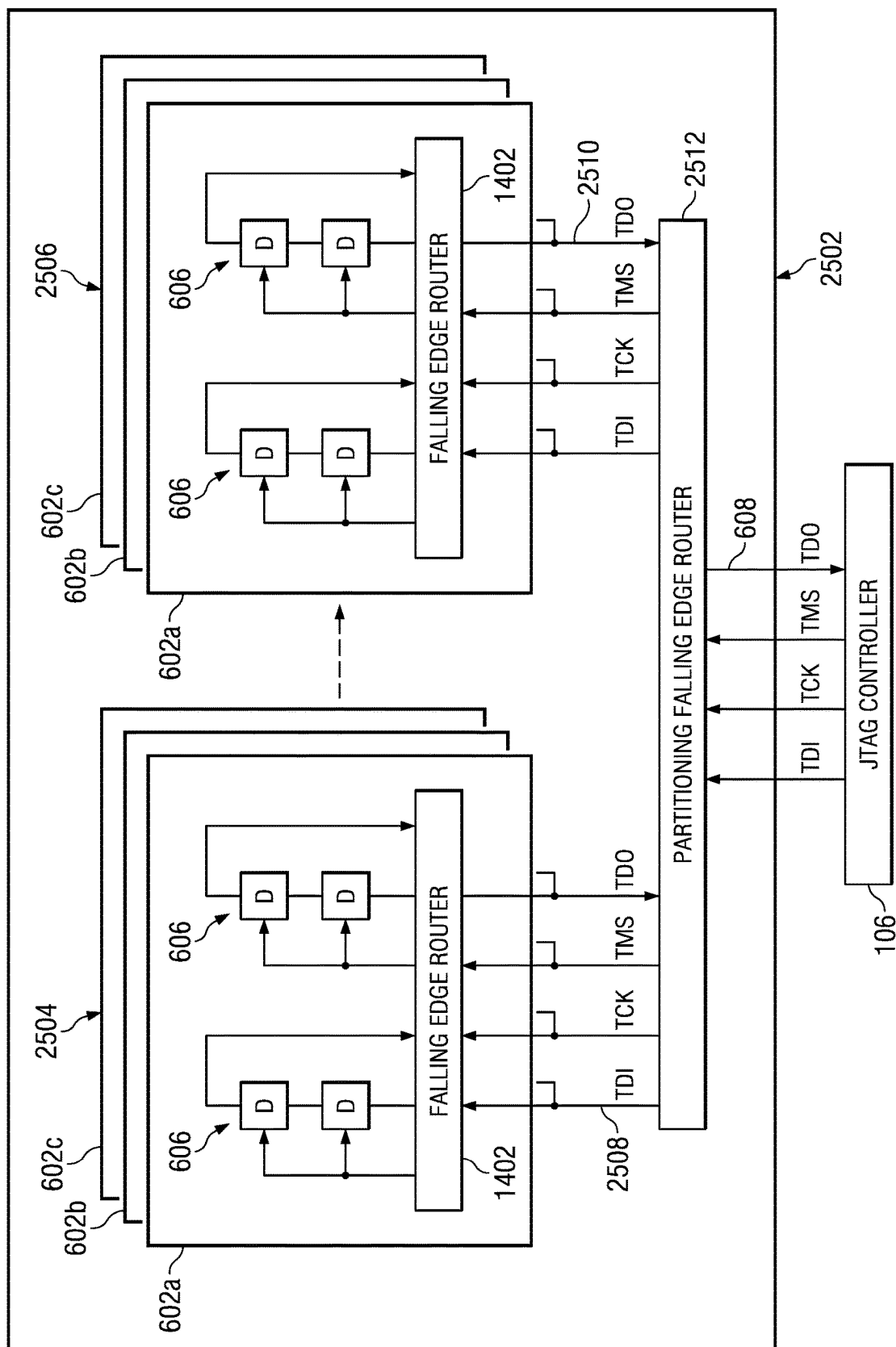


FIG. 25

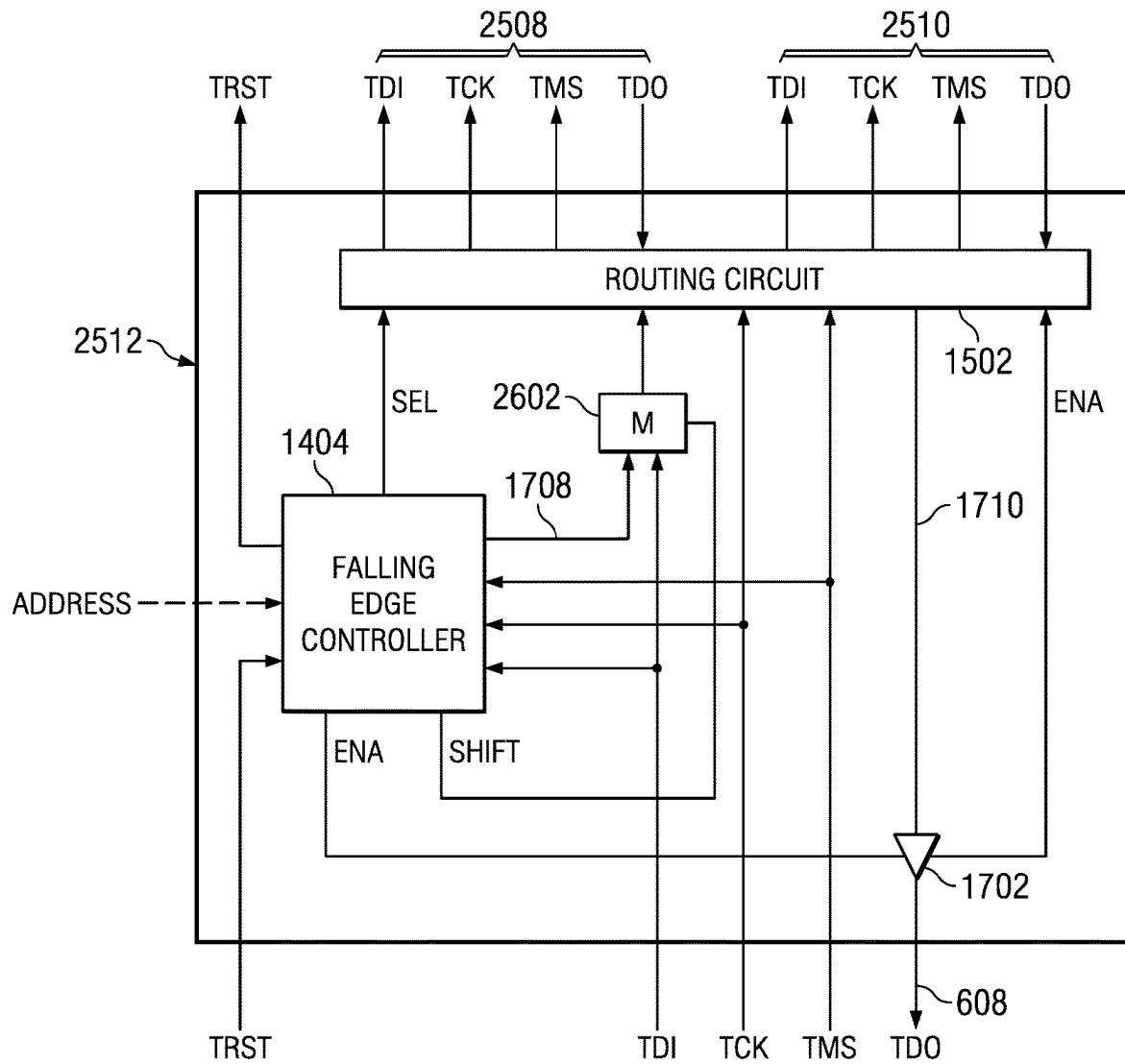
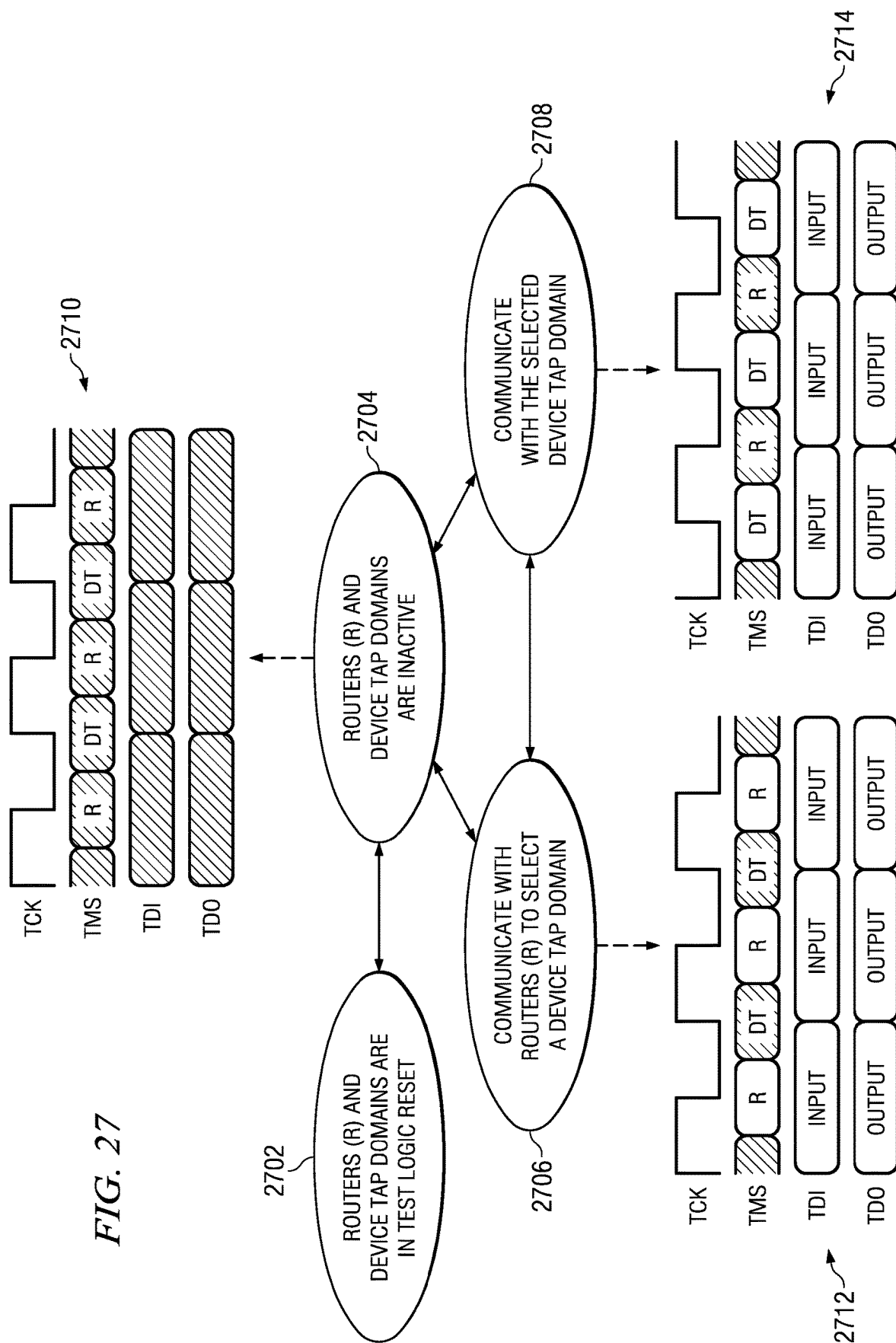
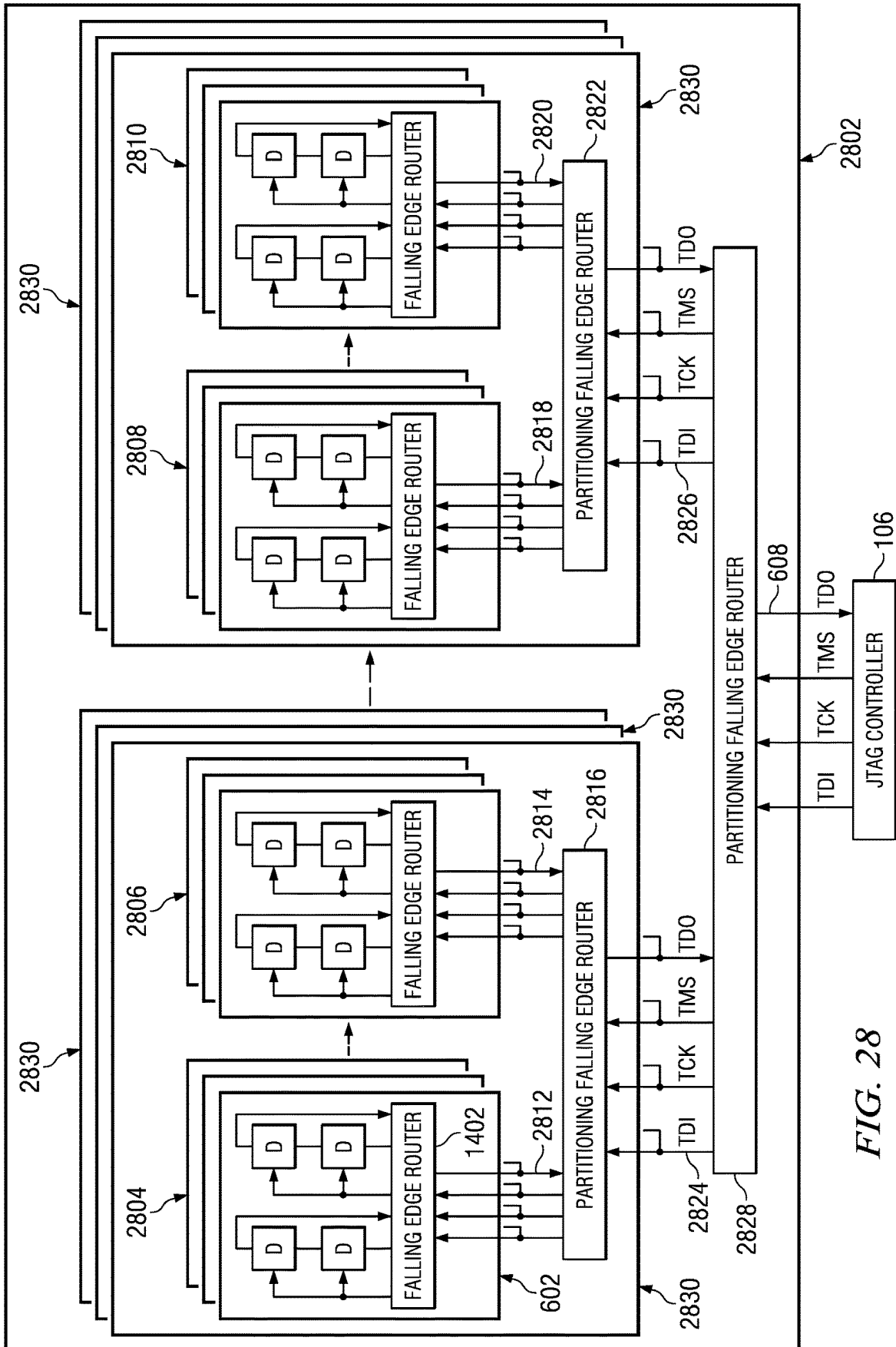
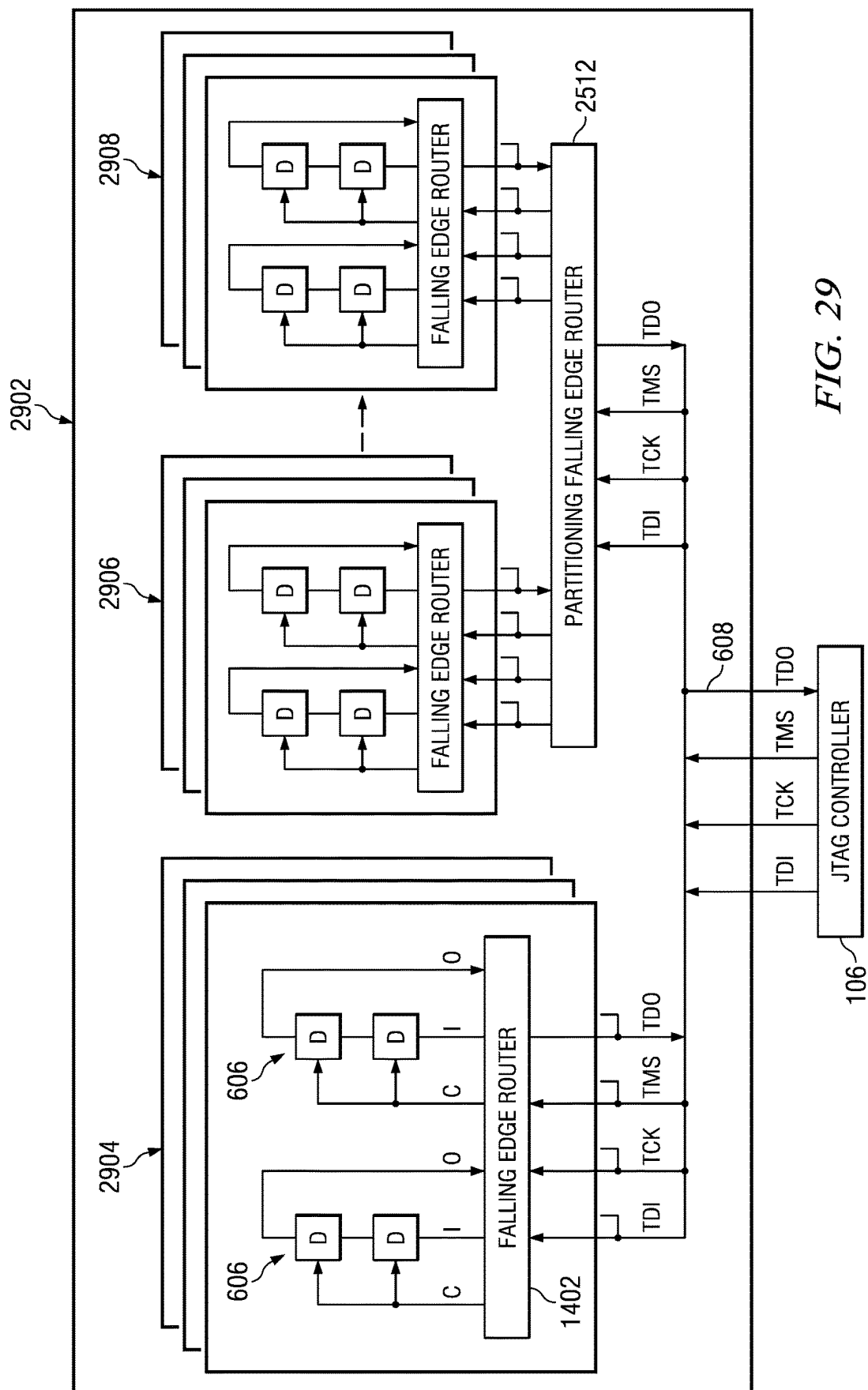


FIG. 26

FIG. 27







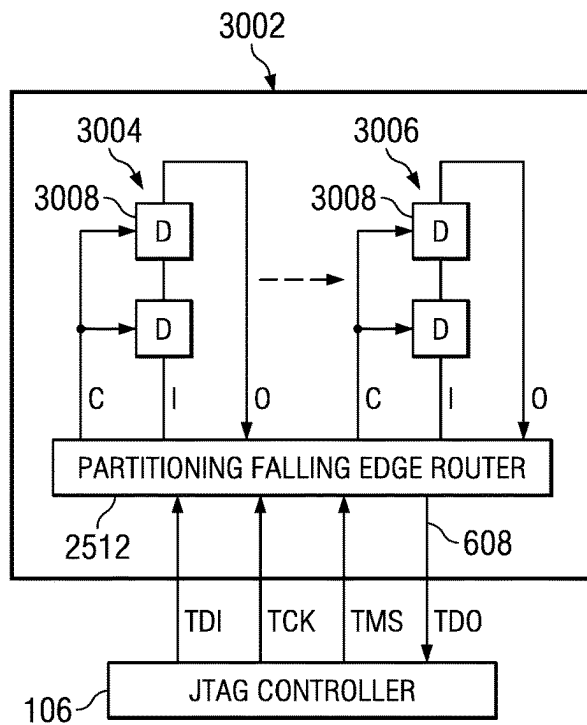


FIG. 30

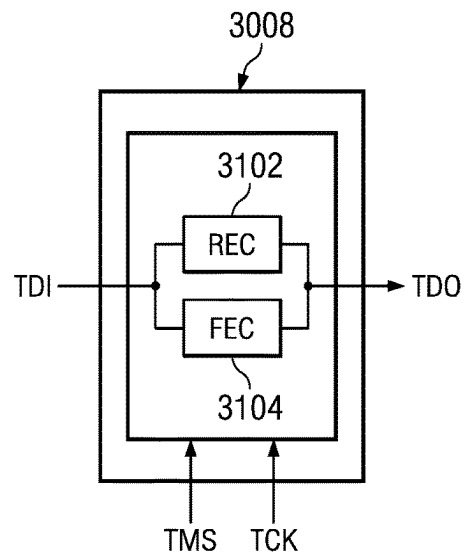


FIG. 31

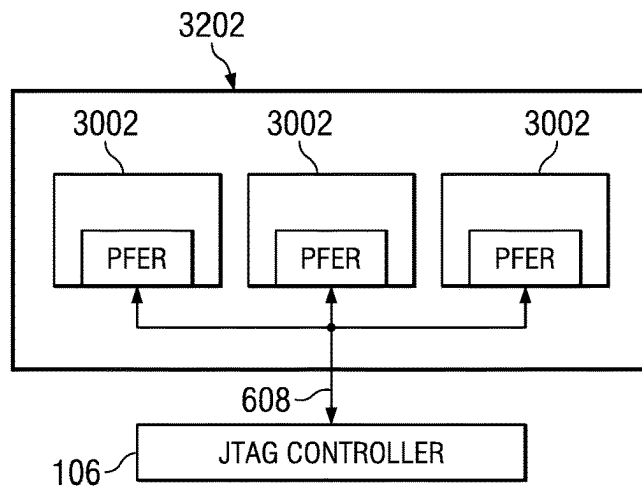


FIG. 32

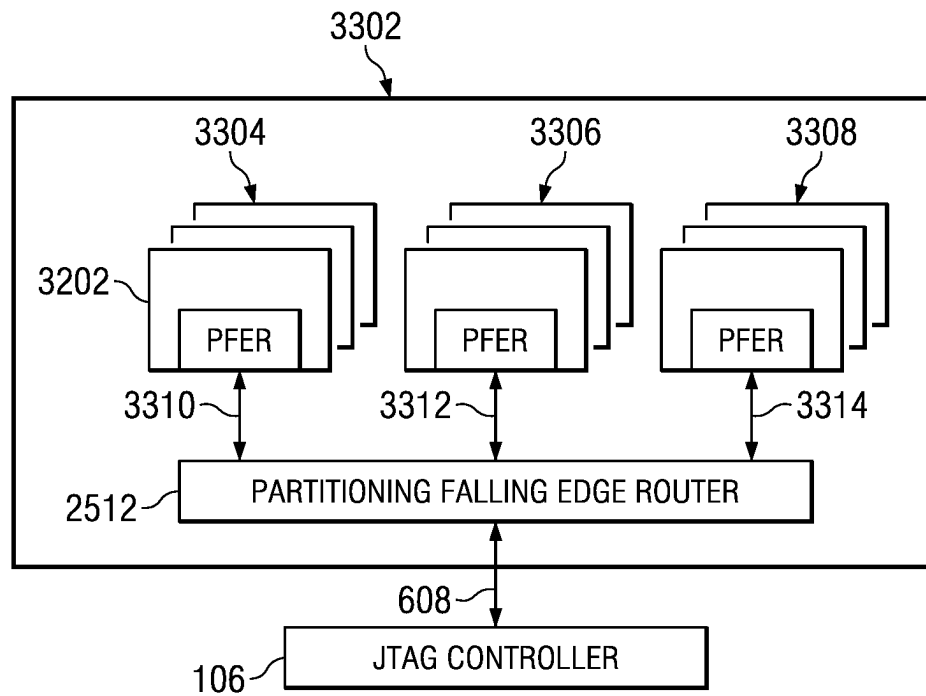


FIG. 33

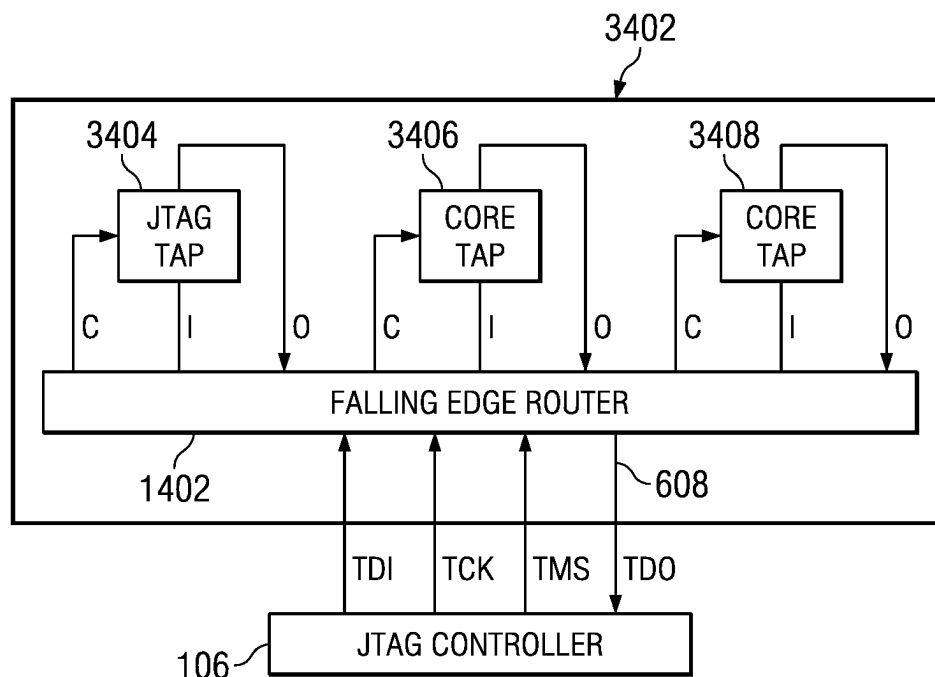
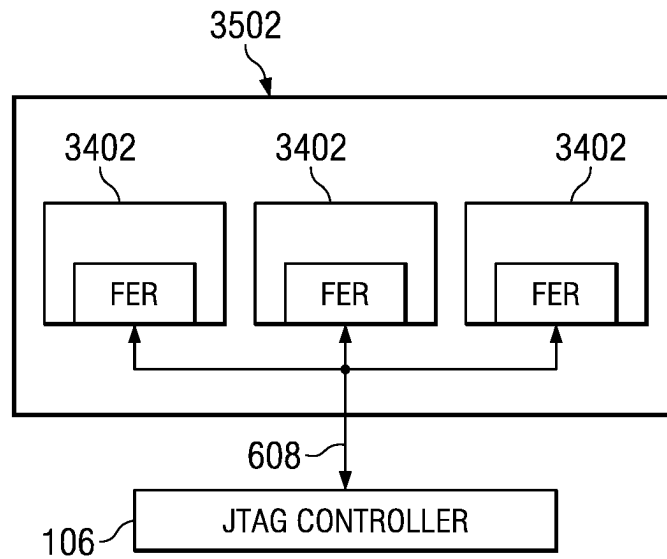
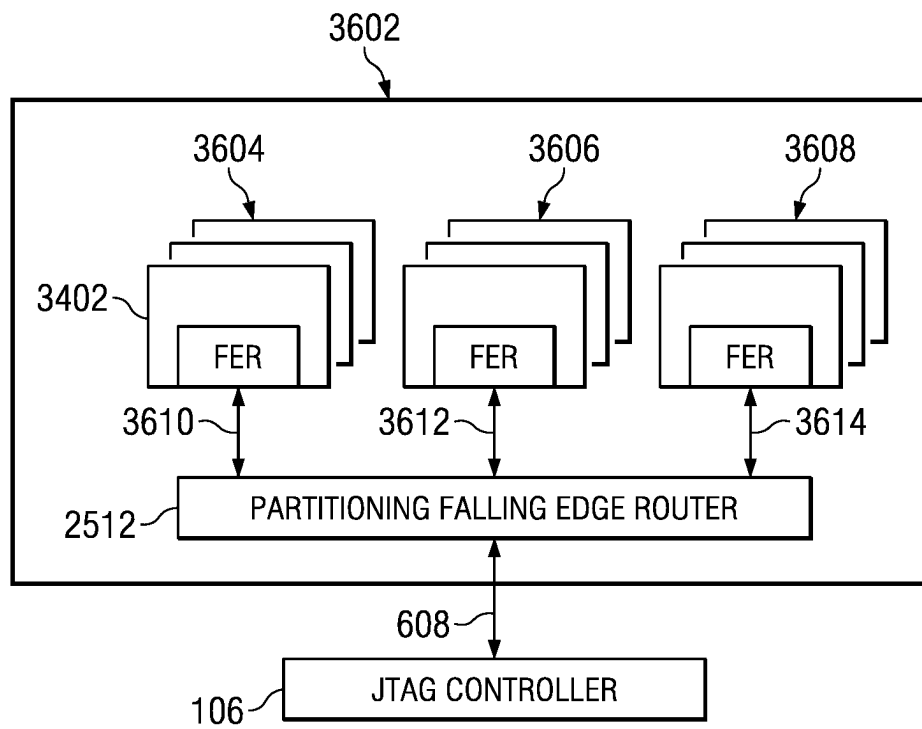


FIG. 34

*FIG. 35**FIG. 36*

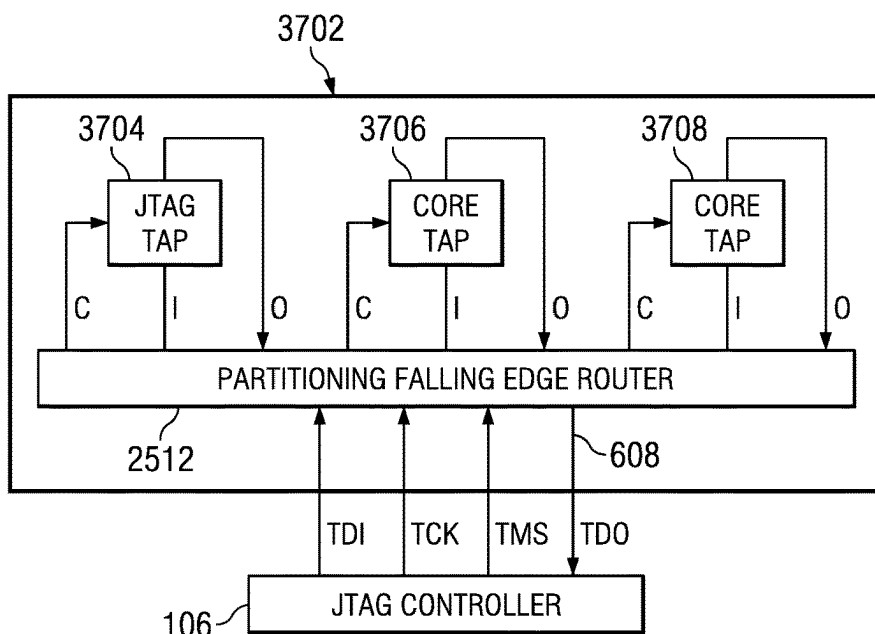


FIG. 37

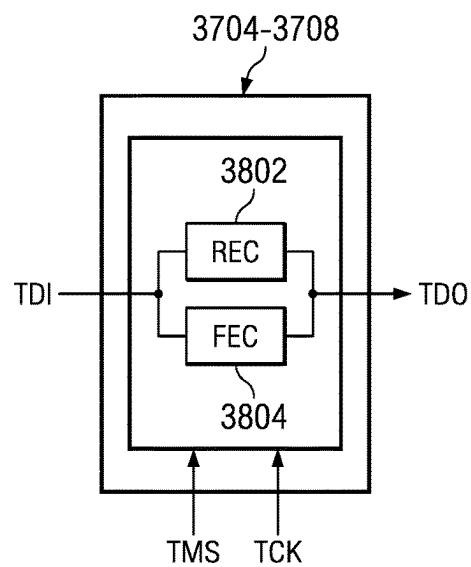


FIG. 38

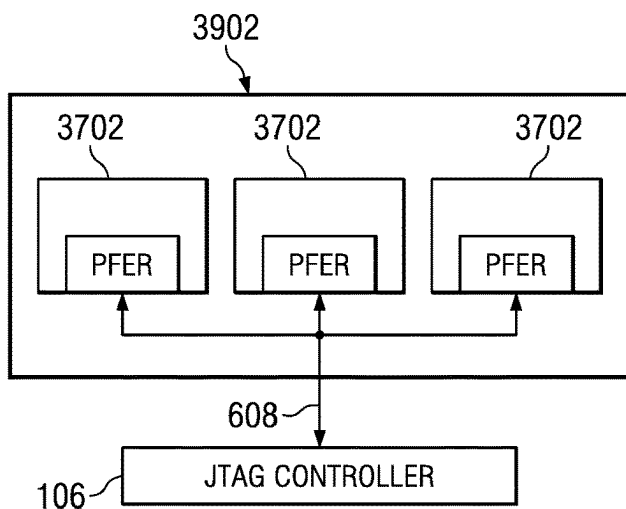


FIG. 39

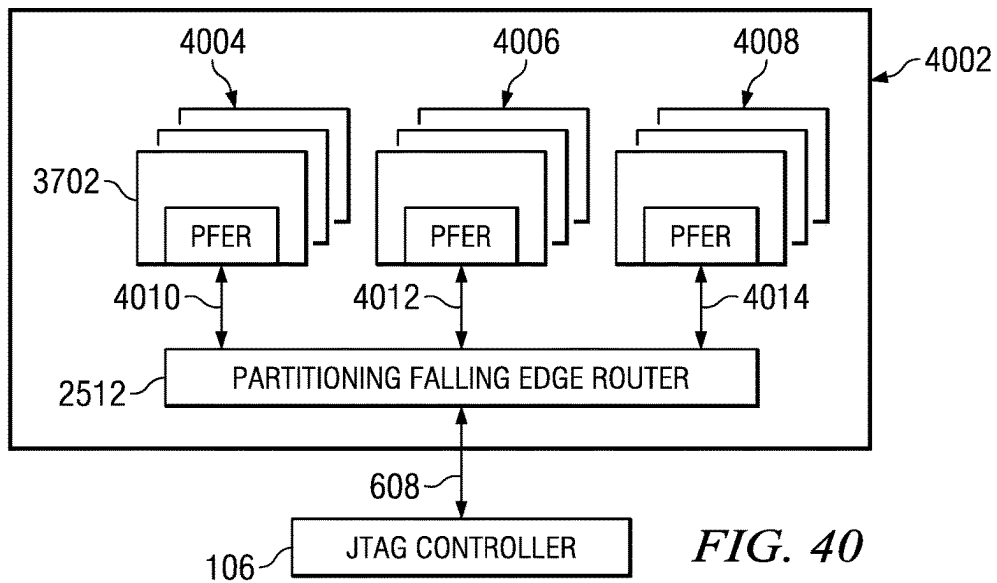


FIG. 40

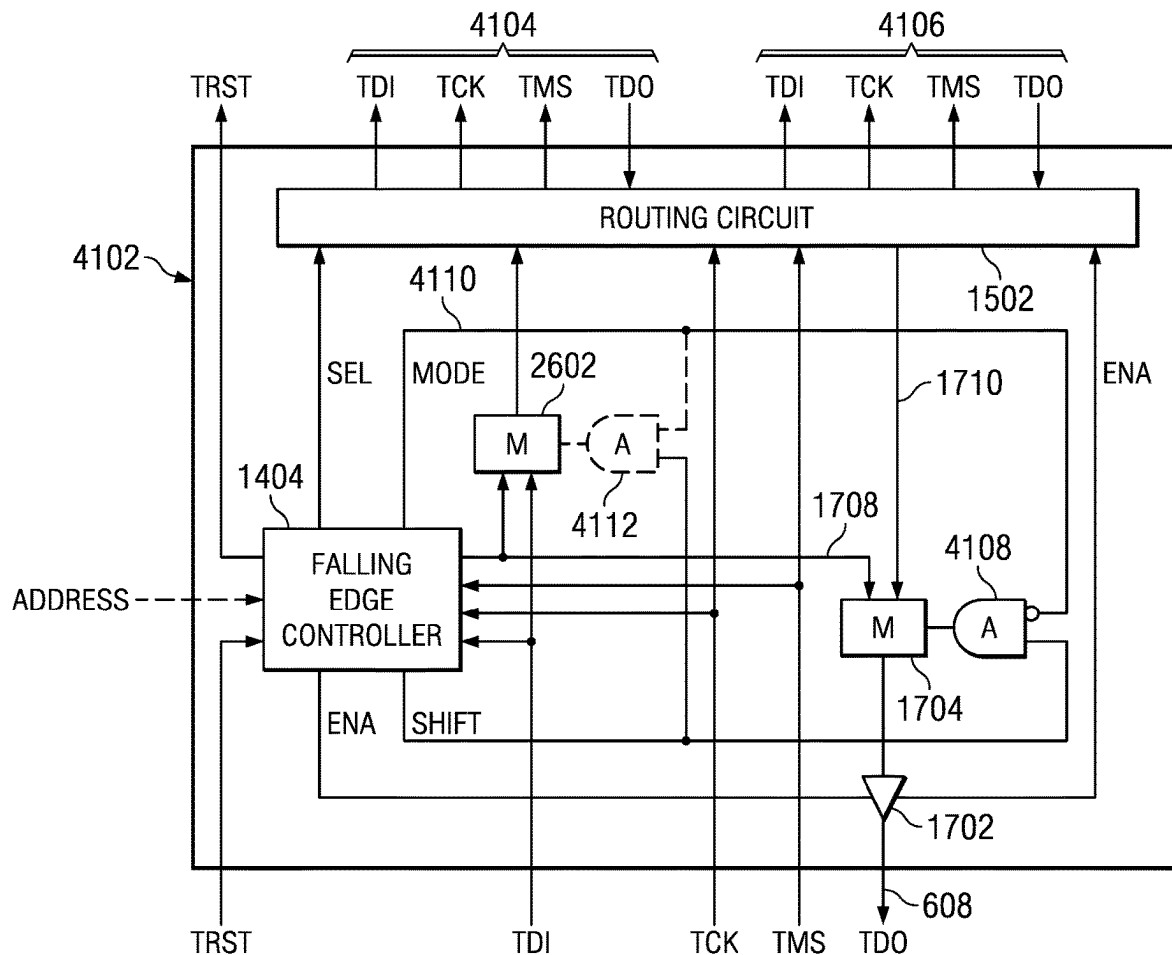
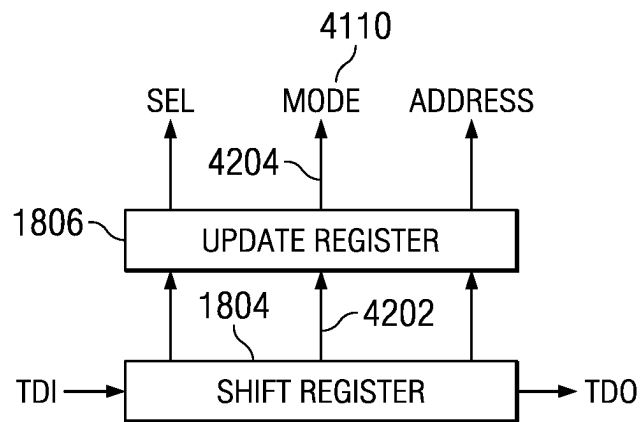
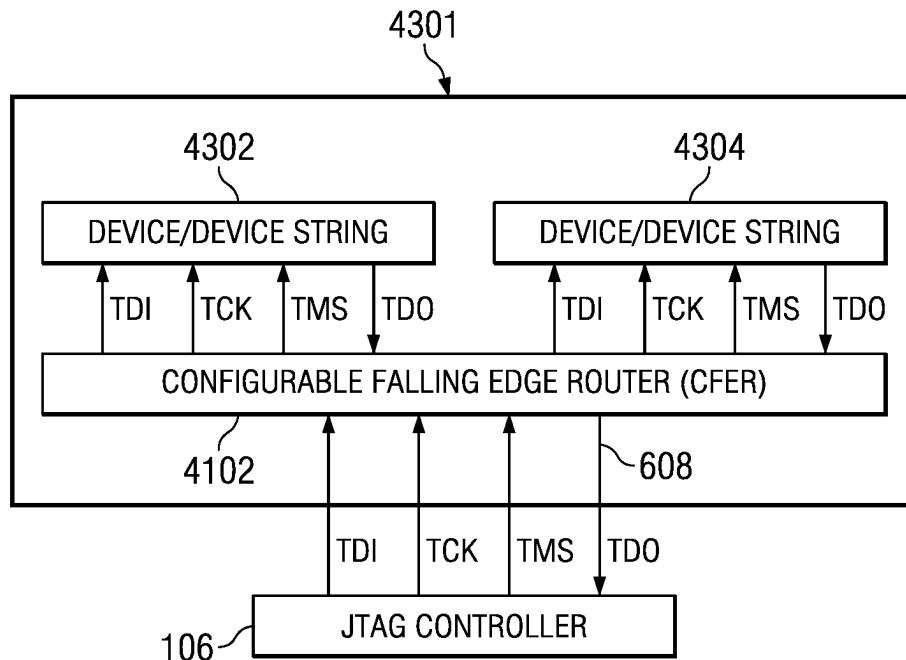


FIG. 41

*FIG. 42**FIG. 43*

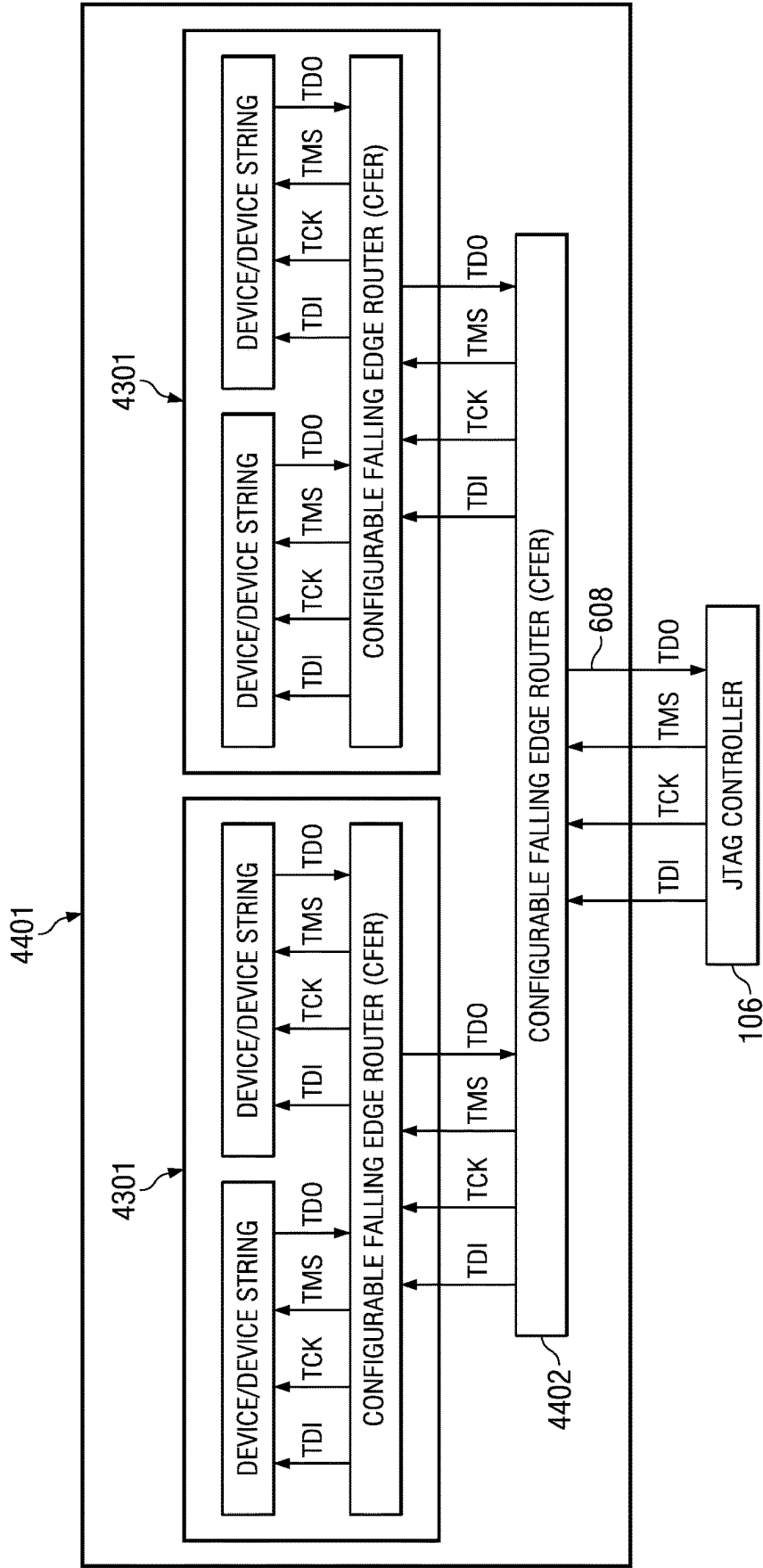


FIG. 44

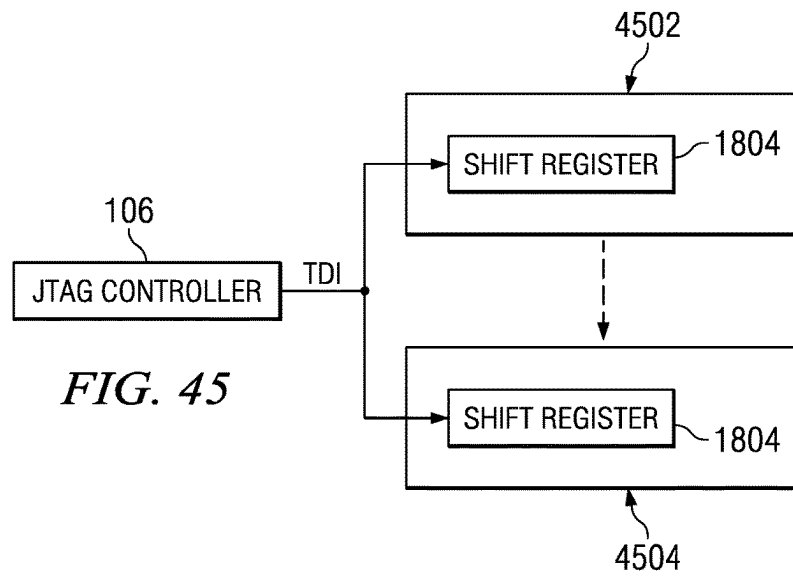


FIG. 45

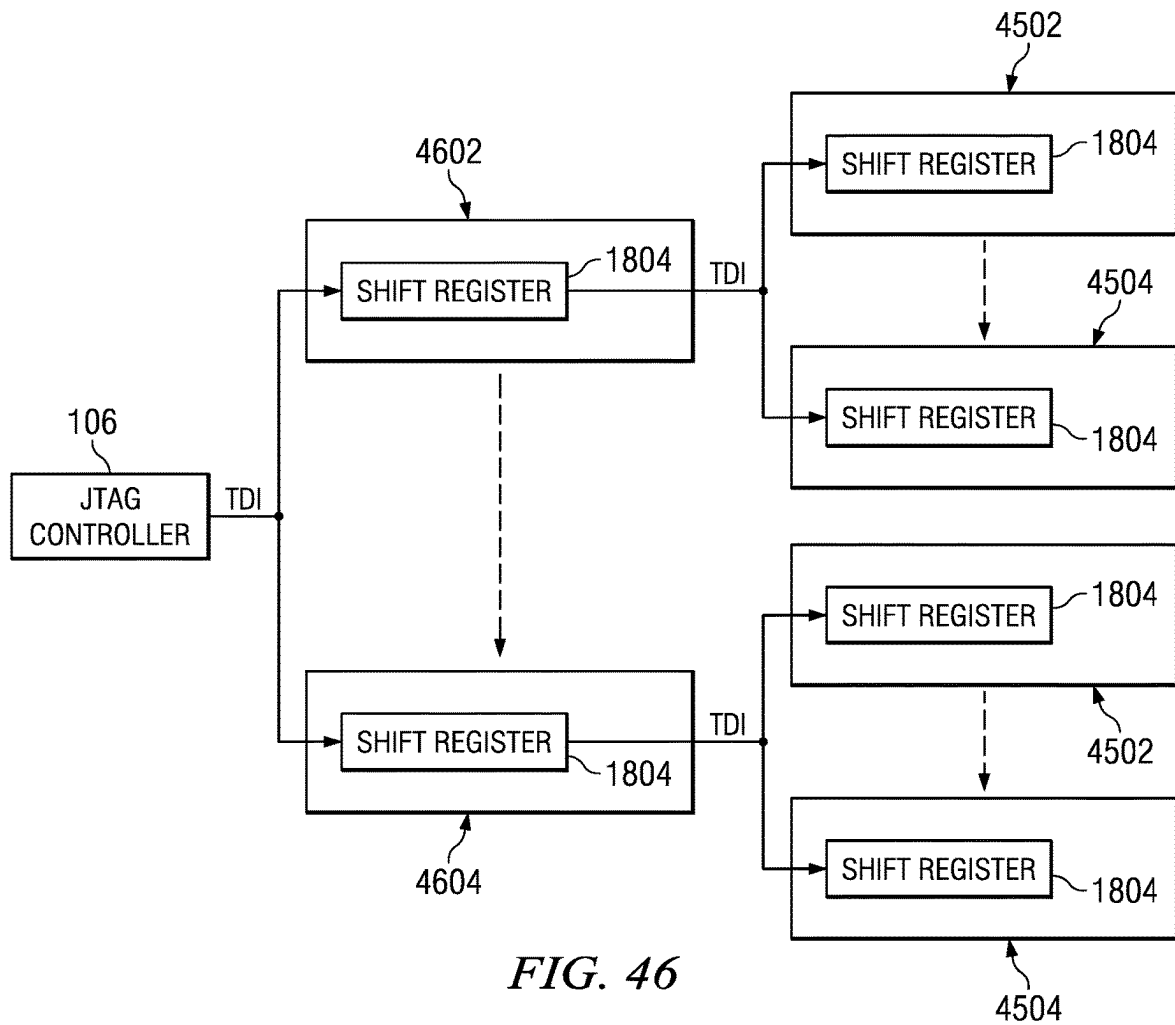


FIG. 46

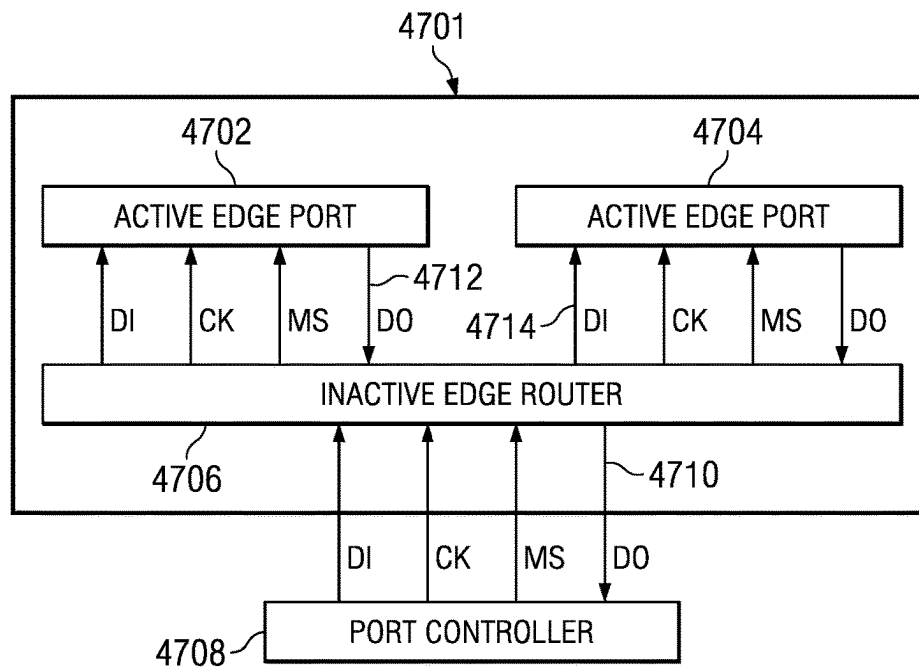


FIG. 47

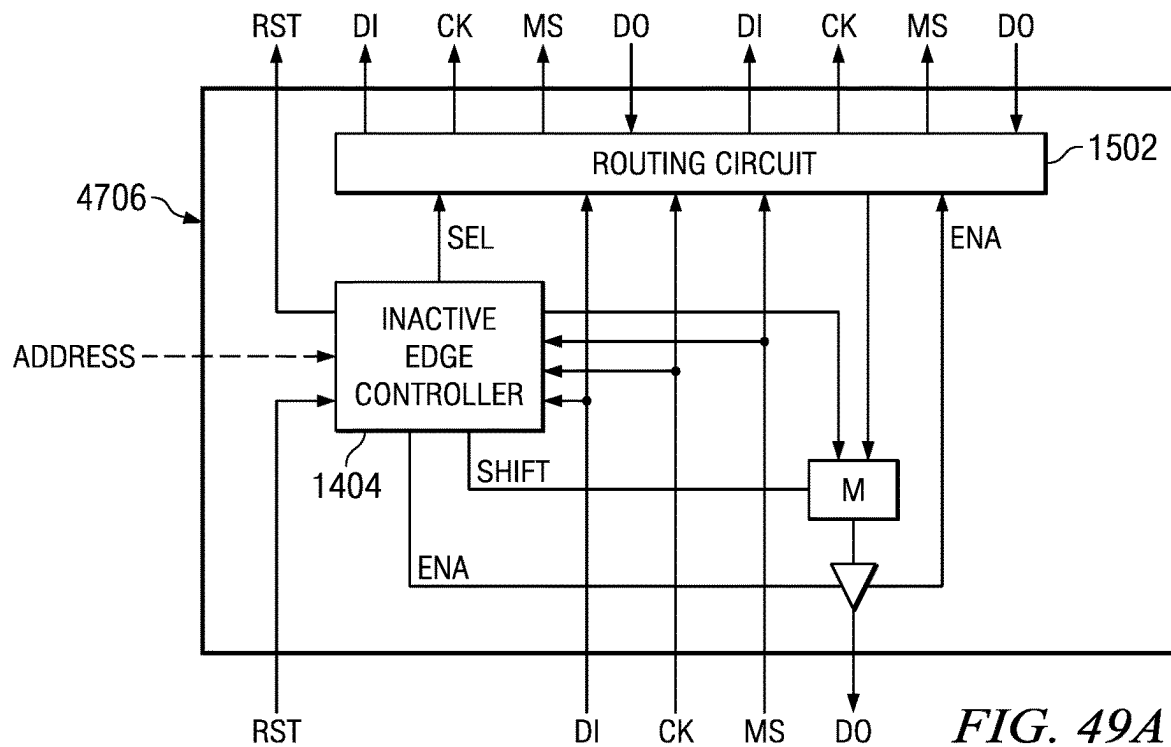


FIG. 49A

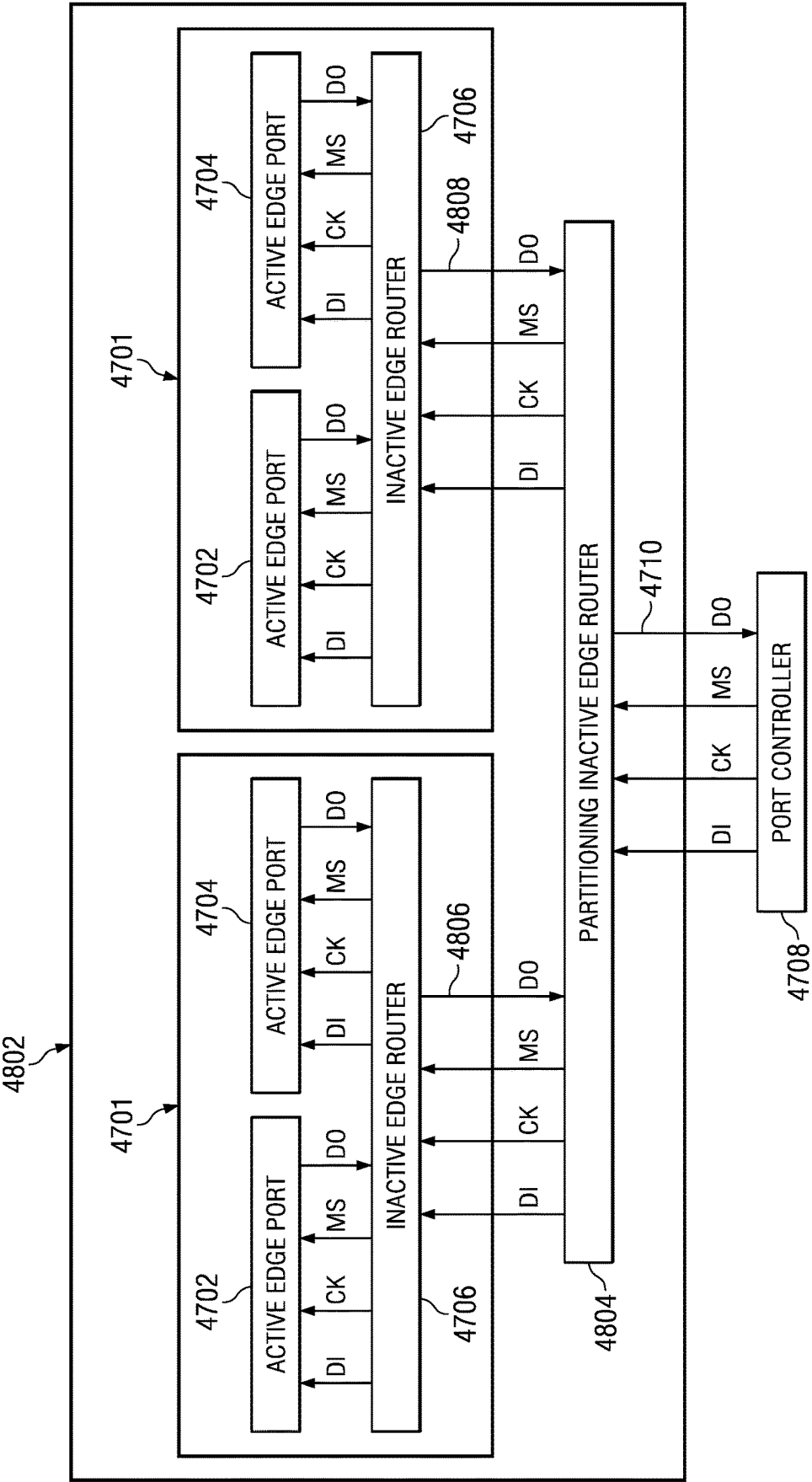


FIG. 48

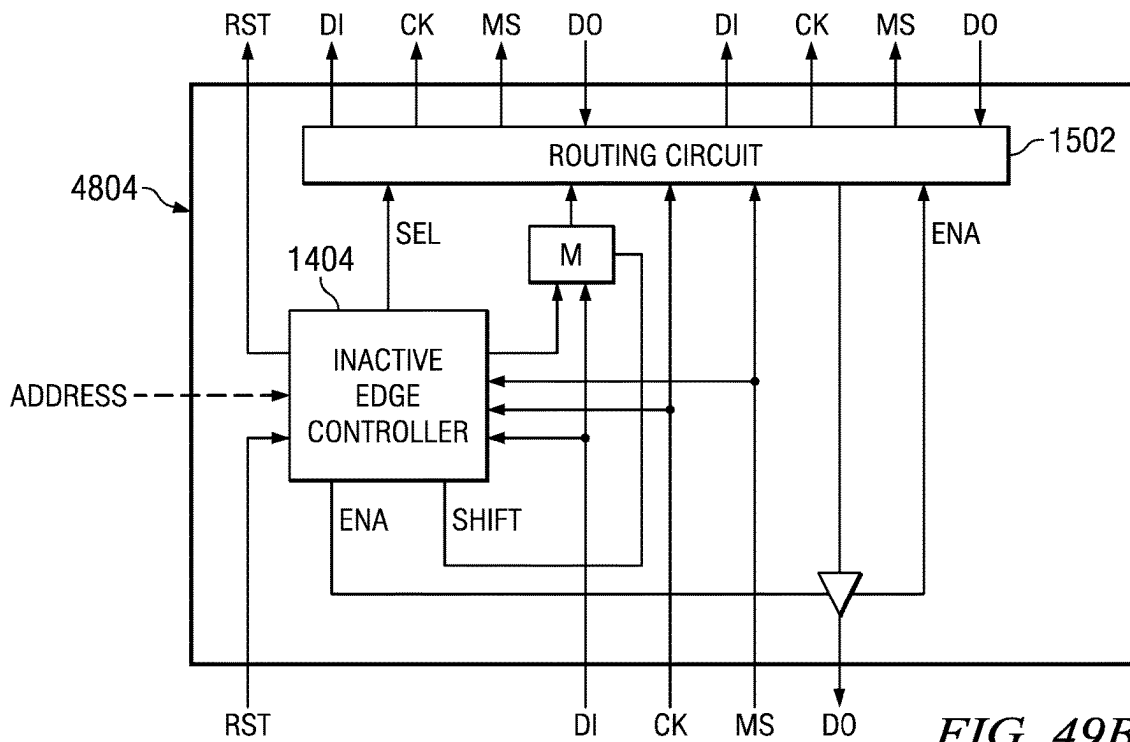


FIG. 49B

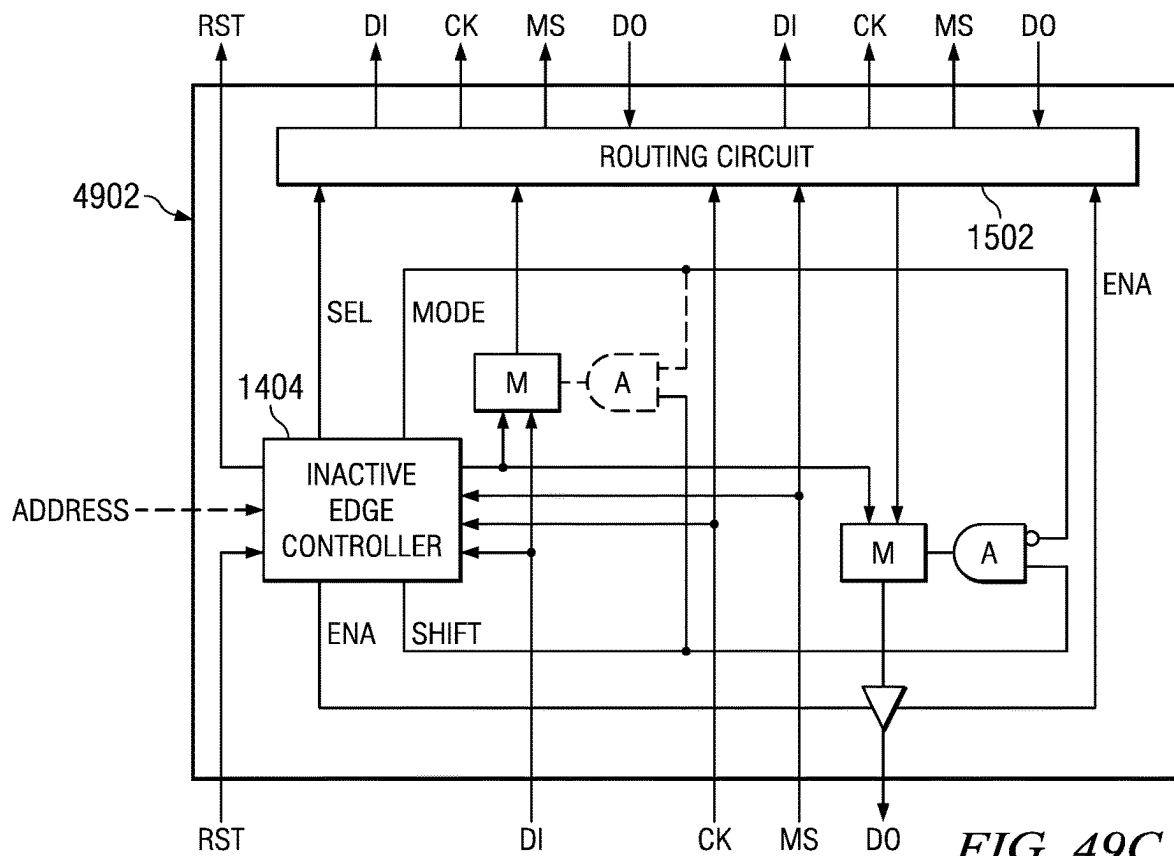


FIG. 49C

FALLING CLOCK EDGE JTAG BUS ROUTERS

CROSS REFERENCE TO RELATED PATENTS

This application is a divisional of application Ser. No. 17/158,438, filed Jan. 26, 2021, currently pending and scheduled to grant as U.S. Pat. No. 11,680,985 on Jun. 20, 2023;

Which was a divisional of prior application Ser. No. 16/776,667, filed Jan. 30, 2020, now U.S. Pat. No. 10,935,601, issued Mar. 2, 2021;

Which was a divisional of prior application Ser. No. 16/228,067, filed Dec. 20, 2018, now U.S. Pat. No. 10,585,144, issued Mar. 10, 2020;

Which was a divisional of prior application Ser. No. 15/499,362, filed Apr. 27, 2017, now U.S. Pat. No. 10,197,628, issued Feb. 5, 2019;

Which was a divisional of prior application Ser. No. 15/179,117, filed Jun. 10, 2016, now U.S. Pat. No. 9,671,463, issued Jun. 6, 2017;

Which was a divisional of prior application Ser. No. 14/054,039, filed Oct. 15, 2013, now U.S. Pat. No. 9,395,412, issued Jul. 19, 2016;

Which was a divisional of prior application Ser. No. 12/968,966, filed Dec. 15, 2010, now U.S. Pat. No. 8,589,714, issued Nov. 19, 2013;

Which claims priority from Provisional Application No. 61/288,055, filed Dec. 18, 2009.

FIELD OF THE DISCLOSURE

This disclosure relates generally to circuits used to route busses in a system and in particular to circuits used to route JTAG (IEEE 1149.1) busses in a system.

BACKGROUND OF THE DISCLOSURE

Most integrated circuit (IC) devices today include a JTAG (Joint Test Action Group) interface comprising TDI (Test Data In), TCK (Test Clock), TMS (Test Mode Select), TRST (Test Reset), and TDO (Test Data Out) bus signal terminals. The JTAG interface on the IC device is used for a myriad of purposes including but not limited to; testing purposes, debugging purposes, and programming purposes.

FIG. 1 illustrates a prior art arrangement of a serial string of IC devices **104** on a board **102**. Each device **104** includes a JTAG interface comprising a control (C) bus of TCK, TMS, and optional TRST signals, an input (I) bus comprising a TDI signal, and an output (O) bus comprising a TDO signal. The control (C) bus (TCK, TMS, TRST), input (I) bus (TDI), and output (O) bus (TDO) are coupled to a JTAG test access port (TAP) **202** in the device as shown in FIG. 2. The TAP **202** is a well known access port defined in IEEE standard 1149.1 that operates to shift instruction and data patterns into and from the device **104** according to the state diagram of FIG. 4 and timing diagram of FIG. 5. In FIG. 5, the rising edge of the TCK signal times the operation of the TAP, the TMS signal controls the state diagram transitions of the TAP, the TDI signal inputs instruction or data patterns to the TAP, and the TDO signal outputs instruction or data patterns from the TAP. The JTAG interface of the devices **104** are connected in series such that the TAPs **202** of all devices **104** can be accessed at the same time from a JTAG controller **106**, via the control (C) bus, input (I) bus, and output (O) bus.

FIG. 3 illustrates a prior art arrangement of boards **102** in a system **302**. The boards typically exist in the backplane of the system **302**. The boards **102** are connected in series such that the TAPs of all devices **104** on each board **102** can be accessed at the same time from a JTAG controller **106**, via the control (C) bus, input (I) bus, and output (O) bus.

As can be seen in FIG. 3, serial arrangements of many boards **102** in a system backplane **302** can problematically extend the access time to the system due to the number of serial bits that must be shifted into and from each board **102** in the system **302** during each JTAG scan operation. Even more problematic, if a board **102** is removed from the system **302** the serial access connection to the JTAG controller **106** is disabled. In response to these system level JTAG scan access problems, JTAG Router devices were developed by National Semiconductor and Texas Instruments that allowed a JTAG controller **106** to directly address and access an individual board **102** in a system **302**.

FIG. 6 illustrates the prior art concept of using a JTAG Router **604** to interface a JTAG controller **106** to one or more JTAG device strings **606** on a board **602**. Each device string **606** may contain one or more devices **104**. In operation, the JTAG controller **106** communicates to the JTAG Router **604** via bus **608** to address it and load selection control to it that selects one or more JTAG device strings **606** for access. The JTAG Router **604** can access one JTAG device string **606** for access or it can serially concatenate and access multiple JTAG device strings **606** together for access. After the addressing and selection control input step, the JTAG controller **106** accesses the selected one or more JTAG device strings **606** via the JTAG Router **604** and bus **608**.

FIG. 6A is provided to indicate that a device string **606** may contain only one device **104**. This will be the case for all device strings **606** shown in this disclosure.

FIG. 7 illustrates a prior art arrangement of boards **602** in a system **702**. The boards **602** typically exist in the backplane of the system **702** so they can be easily removed for replacement or repair. The JTAG interface signals (I, C, O) of each board **602** are connected in parallel such that each board is coupled to the control (C) bus **608** of the JTAG controller, the input (I) bus **608** of the JTAG controller, and the output (O) bus **608** of the JTAG controller. In this arrangement the JTAG controller can individually address and access any board **602** via the board's JTAG Router **604** and bus **608**.

As can be seen in FIG. 7, the problematic scan access time mentioned in regard to serial JTAG access arrangement of FIG. 3 is eliminated since the JTAG controller **106** only performs scan operations to one board **602** at a time, via the board's JTAG Router **604**. Also as seen in FIG. 7, the board removal problem mentioned in regard to FIG. 3 is eliminated since any remaining system boards **602** can be directly addressed and accessed by the JTAG controller **106**, via the board's JTAG Router **604**.

FIG. 8 illustrates a view of a prior art JTAG Router **802** produced by National Semiconductor and referred to as ScanBridge™. The ScanBridge **802** operates to address and select JTAG device strings **606** on boards **602** as described generally in the conceptual JTAG Router descriptions of FIGS. 6 and 7.

FIG. 9 illustrates the ScanBridge circuit **802** in more detail. The ScanBridge includes a Routing Circuit **902** and a JTAG TAP circuit **904**. The JTAG TAP circuit **904** has a first set of TDI, TCK, TMS and TDO signals that are coupled to the JTAG controller bus **608**, a second set of TDI, TCK, TMS, and TDO signals **910** that are coupled to the Routing Circuit **902**, and control (CTL) outputs **912** that are

coupled to the Routing Circuit **902**. The JTAG TAP circuit **904** contains addressing circuitry that can be loaded by the JTAG controller **106** via bus **608** to address the board **602** and routing control circuitry that can be loaded by the JTAG controller via bus **608** to select one or more JTAG device strings **606** on the board for access. In response to the control (CTL) outputs **912** from the JTAG TAP circuit **904**, the Routing Circuit **902** selectively couples the TDI, TCK, TMS, and TDO signals **910** from the JTAG TAP circuit **904** to a selected TDI, TCK, TMS, and TDO signal group **906** that is coupled to a JTAG device string **606**. The Routing Circuit **902** may also concatenate multiple JTAG device strings together and couple them to the TDI, TCK, TMS, and TDO signals **910** from the JTAG TAP circuit **904** in response to the control (CTL) outputs **912** so that they can be accessed together.

FIG. **10** illustrates the process of using JTAG scan operations **1002** to access a selected JTAG device string **606** on a board **602**. Since the ScanBridge's JTAG TAP circuit **904** lies in series between the JTAG controller **106** and the selected JTAG device string **606**, each JTAG scan operation **1002** to the selected JTAG device string **606** must be augmented with instruction and data patterns for the ScanBridge JTAG TAP circuit **904**. Having to augment each JTAG scan operation **1002** with additional instruction and data patterns for the "in series" JTAG TAP circuit **904** is problematic since it lengthens the access time to the selected JTAG device string **606** and requires modifying the existing JTAG pattern set of the devices in the JTAG device string **606**.

FIG. **11** illustrates a view of a prior art JTAG Router **1102** produced by Texas Instruments and referred to as a linking Addressable Scan Port (ASP). The ASP **1102** operates to address and select JTAG device strings **606** on boards **602** as described generally in the conceptual JTAG Router descriptions of FIGS. **6** and **7**.

FIG. **12** illustrates the ASP circuit **1102** in more detail. The ASP includes a Routing Circuit **1202** and a Shadow Protocol Controller **1204**. The Shadow Protocol Controller **1204** has a set of TDI, TCK, TMS and TDO signals that are coupled to the JTAG controller bus **608**, and control (CTL) outputs **1206** that are coupled to the Routing Circuit **1202**. The Shadow Protocol Controller **1204** contains addressing circuitry that can be loaded by the JTAG controller **106** via bus **608** to address the board **602** and routing control circuitry that can be loaded by the JTAG controller via bus **608** to select one or more JTAG device strings **606** on the board for access. In response to the control (CTL) outputs **1206** from the Shadow Protocol Controller **1204**, the Routing Circuit **1202** selectively couples the TDI, TCK, TMS, and TDO signals from the JTAG controller **106** to a selected TDI, TCK, TMS, and TDO signal group **906** that is coupled to a JTAG device string **606**. The Routing Circuit **1202** may also concatenate multiple JTAG device strings together and couple them to the TDI, TCK, TMS, and TDO signals via bus **608** from the JTAG controller **106** in response to the control (CTL) outputs **1206** so that they can be accessed together.

As seen, the Shadow Protocol Controller **1204** does not exist in series in bus **608** between the JTAG controller **106** and the Routing Circuit **1202**, but is simply coupled to bus **608**. The JTAG controller **106** communicates to the Shadow Protocol Controller **1204** using Shadow Protocol Messages to load board address and device string selection information during times when JTAG bus operations are inactive in the Run Test/Idle, Pause-DR and Pause-IR states of FIG. **4**.

FIG. **13** illustrates the process of using Shadow Protocol Messages **1302** and **1304** to access a selected JTAG device string **606** on a board **602**. When JTAG bus operations are inactive in one of the states mentioned above, the JTAG controller **106** inputs a Shadow Protocol Message request **1302** to the Shadow Protocol Controller **1204** that contains the board address and device string selection information. In response to the request **1302**, the Shadow Protocol Controller **1204** outputs a Shadow Protocol Message acknowledge **1304** to the JTAG controller **106** to confirm the address and selection information, then connects the selected device string **606** to the JTAG controller **106** via bus **608**. Following the connect operation, the JTAG controller **106** performs JTAG scan operations **1306** to access the selected device string **606**. As can be seen, the JTAG scan operations **1306** only include instruction and data patterns required by the devices in the selected device string **606**. Thus the ASP **1102** does not lengthen the access time to the selected JTAG device string **606** and does not require modifying the existing JTAG pattern set of the devices in the JTAG device string **606**, as does the ScanBridge **802**. However, the Shadow Protocol Messages **1302** and **1304** are based on Manchester-like encoding and decoding, which requires the Shadow Protocol Controller **1204** to be fairly complex which adds to the cost of the ASP device **1102**.

BRIEF SUMMARY OF THE DISCLOSURE

This disclosure describes a method and apparatus for allowing a JTAG controller to access JTAG device strings on a board or other substrate using a simplified JTAG Router device that operates on the falling edge of the JTAG TCK signal.

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWINGS

FIG. **1** illustrates a conventional arrangement of IC devices on a board connected to a JTAG controller.

FIG. **2** illustrate a conventional IC device having a JTAG TAP interface.

FIG. **3** illustrates a conventional arrangement of boards in a system serially connected to a JTAG controller.

FIG. **4** illustrates a conventional JTAG TAP state diagram.

FIG. **5** illustrates a timing diagram of a conventional JTAG interface.

FIG. **6** illustrates a conventional arrangement of strings of IC devices on a board connectable to a JTAG controller via a JTAG router.

FIG. **6A** illustrates a conventional arrangement of a string of one IC Device on a board connectable to a JTAG controller via a JTAG router.

FIG. **7** illustrates a conventional arrangement of boards in a system, each being selectively connected to a JTAG controller.

FIG. **8** illustrates a conventional arrangement of strings of one or more IC devices on a board connectable to a JTAG controller via a National Semiconductor ScanBridge™ JTAG router.

FIG. **9** illustrates a view of the National Semiconductor ScanBridge™ JTAG router.

FIG. **10** illustrates the operation of the National Semiconductor ScanBridge™ JTAG router.

FIG. **11** illustrates a conventional arrangement of strings of one or more IC devices on a board connectable to a JTAG controller via a Texas Instruments Addressable Scan Port (ASP) JTAG router.

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FIG. 12 illustrates a view of the Texas Instruments Addressable Scan Port (ASP) JTAG router.

FIG. 13 illustrates the operation of the Texas Instruments Addressable Scan Port (ASP) JTAG router.

FIG. 14 illustrates an arrangement of strings of one or more IC devices on a board connectable to a JTAG controller via the Falling Edge Router (FER) according to the disclosure.

FIG. 15 illustrates a view of the Falling Edge Router according to the disclosure.

FIG. 16 illustrates the operation of the Falling Edge Router according to the disclosure.

FIG. 17 illustrates one preferred, but not limited to, example embodiment of the Falling Edge Router of FIG. 16 according to the disclosure.

FIG. 18 illustrates one preferred, but not limited to, embodiment of the Falling Edge Controller of the FIG. 17 Falling Edge Router according to the disclosure.

FIG. 19 illustrates one preferred, but not limited to, example embodiment of the Address Circuit of the Falling Edge Controller of FIG. 18 according to the disclosure.

FIG. 20 illustrates one preferred, but not limited to, example of the state diagram of the Controller of the Falling Edge Controller of FIG. 18 according to the disclosure.

FIG. 21 illustrates one preferred, but not limited to, example embodiment of the Routing Circuit of the Falling Edge Controller of FIG. 17 according to the disclosure.

FIG. 22 illustrates the operation of the decode circuit 2116 in response to the SEL and ENA signals from Falling Edge Controller 1404.

FIG. 23 illustrates one preferred, but not limited to, example of the operational states and timing of the Routers and TAP domains of FIG. 17 according to the disclosure.

FIG. 24 illustrates a system comprising Falling Edge Router equipped sub-systems coupled to a JTAG controller according to the disclosure.

FIG. 25 illustrates a system comprising groups of Falling Edge Router equipped sub-systems coupled to a JTAG controller via a Partitioning Falling Edge Router according to the disclosure.

FIG. 26 illustrates one preferred, but not limited to, example embodiment of the Partitioning Falling Edge Router of FIG. 25 according to the disclosure.

FIG. 27 illustrates one preferred, but not limited to, example of the operational states and timing of the Routers and TAP domains of FIG. 25 according to the disclosure.

FIG. 28 illustrates a system comprising groups of one or more Falling Edge Router equipped sub-systems coupled to a JTAG controller via a hierarchy of Partitioning Falling Edge Routers according to the disclosure.

FIG. 29 illustrates a system comprising groups of one or more Falling Edge Router equipped sub-systems coupled to a JTAG controller either directly or via an intervening Partitioning Falling Edge Router according to the disclosure.

FIG. 30 illustrates a system comprising strings of one or more devices each containing Rising and Falling Edge Circuitry coupled to a JTAG controller via a Partitioning Falling Edge Router according to the disclosure.

FIG. 31 illustrates an example of a device containing Rising and Falling Edge Circuitry according to the disclosure.

FIG. 32 illustrates a system comprising multiple FIG. 30 systems coupled to a JTAG controller according to the disclosure.

FIG. 33 illustrates a system comprising multiple groups of FIG. 30 systems coupled to a JTAG controller via a Partitioning Falling Edge Controller according to the disclosure.

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FIG. 34 illustrates a device comprising a JTAG TAP domain, a first core TAP domain, and a second core TAP domain coupled to a JTAG controller via a Falling Edge Router according to the disclosure.

FIG. 35 illustrates a system comprising multiple FIG. 34 devices coupled to a JTAG controller according to the disclosure.

FIG. 36 illustrates a system comprising multiple groups of FIG. 34 devices coupled to a JTAG controller via a Partitioning Falling Edge Router according to the disclosure.

FIG. 37 illustrates a device comprising a JTAG TAP domain, a first core TAP domain, and a second core TAP domain coupled to a JTAG controller via a Partitioning Falling Edge Router according to the disclosure.

FIG. 38 illustrates an example of a modified TAP circuit domain containing Rising and Falling Edge Circuitry according to the disclosure.

FIG. 39 illustrates a system comprising multiple FIG. 37 devices coupled to a JTAG controller according to the disclosure.

FIG. 40 illustrates a system comprising multiple groups of FIG. 37 devices coupled to a JTAG controller via a Partitioning Falling Edge Router according to the disclosure.

FIG. 41 illustrates a view of a Configurable Falling Edge Router according to the disclosure.

FIG. 42 illustrates the shift and update registers of the Falling Edge Controller of FIG. 41 according to the disclosure.

FIG. 43 illustrates a system comprising multiple devices or device strings coupled to a JTAG controller via the Configurable Falling Edge Router of FIG. 41 according to the disclosure.

FIG. 44 illustrates a further system comprising multiple FIG. 43 systems coupled to a JTAG controller via the Configurable Falling Edge Router of FIG. 41 according to the disclosure.

FIG. 45 illustrates equal length shift registers of multiple Falling Edge Routers coupled to the TDI signal output from a JTAG controller according to the disclosure.

FIG. 46 illustrates equal length shift registers of multiple Partitioning Falling Edge Routers and multiple Falling Edge Routers coupled to the TDI signal output from a JTAG controller according to the disclosure.

FIG. 47 illustrates a system comprising multiple Active Edge Ports coupled to a Port controller via an Inactive Edge Router according to the disclosure.

FIG. 48 illustrates a further system comprising multiple FIG. 47 systems coupled to a Port controller via a Partitioning Inactive Edge Router according to the disclosure.

FIG. 49A illustrates an example implementation of an Inactive Edge Router according to the disclosure.

FIG. 49B illustrates an example implementation of a Partitioning Inactive Edge Router according to the disclosure.

FIG. 49C illustrates an example implementation of a Configurable Inactive Edge Router according to the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

FIG. 14 illustrates a view of a falling edge bus router 1402, hereafter referred to as a Falling Edge Router (FER), that operates to couple a JTAG controller 106 to a string 606 of one or more IC devices (D) on a board 602, according to

the present disclosure. The FER **1402** operates on the falling edge of the TCK to address and select IC device strings **606** on the board **602**.

It should be understood that while the board **602** of FIG. **14** is described as being a board with IC device strings **606**, the board **602** could also be an IC with strings **606** of one or more embedded core circuit devices or a core circuit with strings **606** of one or more further embedded core circuit devices. This will be the case for other similar example figures in this disclosure.

FIG. **15** illustrates the FER circuit **1402** in more detail. The FER includes a Routing Circuit **1502** and a Falling Edge Controller **1404**. The Falling Edge Controller **1404** has a set of TDI, TCK, TMS and TDO signals that are coupled to the JTAG controller bus **608**, and control (CTL) outputs **1406** that are coupled to the Routing Circuit **1502**. The Falling Edge Controller **1404** contains addressing circuitry that can be loaded by the JTAG controller **106** via bus **608** to address the board **602** and routing control circuitry that can be loaded by the JTAG controller **106** via bus **608** to select one or more JTAG device strings **606** on the board **602** for access. In response to the control (CTL) outputs **1406** from the Falling Edge Controller **1404**, the Routing Circuit **1502** selectively couples the TDI, TCK, TMS, and TDO signals from the JTAG controller **106** to a selected TDI, TCK, TMS, and TDO signal group **906** that is coupled to a JTAG device string **606**. The Routing Circuit **1502** may also concatenate multiple JTAG device strings together and couple them to the TDI, TCK, TMS, and TDO signals from the JTAG controller **106** in response to the control (CTL) outputs **1406** so that they can be accessed together.

As seen, the Falling Edge Controller **1404**, like the Shadow Protocol Controller **1204** of ASP **1102**, does not exist in series in bus **608** between the JTAG controller **106** and the Routing Circuit **1402**, but is simply coupled to bus **608**. The JTAG controller **106** communicates to the Falling Edge Controller **1404** using simple falling TCK edge scan operations to load board address and device string selection information during times when JTAG bus operations are inactive in the Run Test/Idle, Pause-DR and Pause-IR states of FIG. **4**.

FIG. **16** illustrates the process of using a falling TCK edge scan operation **1602** from a JTAG controller **106** to access a selected JTAG device string **606** on a board **602**. When JTAG bus operations are inactive in one of the states mentioned above, the JTAG controller **106** performs the falling edge scan operation **1602** to input board address and device string selection information to the Falling Edge Controller **1404** of FER **1402**. In response to the falling edge scan operation **1602**, the Falling Edge Controller **1404** outputs control (CTL) **1406** to routing circuit **1502** of FER **1402** to connect the selected device string **606** to the JTAG controller **106** via bus **608**. Following the connect operation, the JTAG controller **106** performs rising TCK edge JTAG scan operations **1604** to access the selected device string **606**.

As can be seen, the rising TCK edge JTAG scan operations **1604** only include instruction and data patterns required by the devices in the selected device string **606**. Thus the FER **1402**, like the ASP **1102**, does not lengthen the access time to the selected JTAG device string **606** and does not require modifying the existing JTAG pattern set of the devices in the JTAG device string **606**, as does the Scan-Bridge **802**. Furthermore, since the FER **1402** uses simple falling TCK edge scan operations to address and select a device string **606**, the Falling Edge Controller **1404** of the FER **1402** is an extremely simple circuit compared to the

ASP's Shadow Protocol Controller **1204**, which reduces the cost of the FER device **1402**.

FIG. **17** illustrates one example implementation of FER **1402** according to the present disclosure. However, the disclosure is not limited to only this one implementation example of FER **1402**. In this one example, the FER comprises the Falling Edge Controller **1404**, the Routing Circuit **1502**, a TDO output buffer **1702**, and a multiplexer **1704**, all connected as shown.

The Falling Edge Controller **1404** has inputs for inputting the TDI, TCK, TMS and optional TRST signals from a JTAG controller **106**, via bus **608**, and optional external Address signals. The Falling Edge Controller **1404** has outputs for outputting a TDO signal **1708** to multiplexer **1704**, a Shift signal to multiplexer **1704**, an enable (ENA) signal to TDO buffer **1702** and Routing Circuit **1502**, select (SEL) signals to Routing Circuit **1502**, and a TRST output signal to devices in JTAG device strings **606**. The ENA and SEL signals on control (CTL) bus **1406** of FIG. **15**.

The Routing Circuit **1402** has inputs for inputting the TDI, TCK, and TMS signals from bus **608**, the SEL and ENA signals from the Falling Edge Controller **1404**, a TDOa signal of a bus **906** of a first device string **606**, and a TDOb signal of a bus **906** of a second device string **606**. The Routing Circuit has outputs for outputting TDIa, TCKa, TMSa signals of the bus **906** of the first device sting **606**, TDIb, TCKb, TMSb signals of the bus **906** of the second device sting **906**, and a TDO signal **1710** to multiplexer **1704**.

Multiplexer **1704** inputs TDO signals **1708** and **1710** and outputs the selected TDO signal to the TDO signal of bus **608** via buffer **1702**. The Shift input controls which TDO signal **1708** or **1710** is selected to be output on the TDO signal of bus **608**.

While JTAG operations on bus **608** are inactive in the TAP states mentioned above, the Falling Edge Controller **1404** responds to the TMS input on the falling edge of the TCK input to input new address and device string select information on the TDI input and output existing address and device string select information on the TDO signal **1708**. If the address data input on TDI matches the address of the Falling Edge Controller **1404**, the ENA signal is asserted to enable the TDO output buffer **1702** and the Routing Circuit **1402**. While the ENA signal is asserted, the Routing Circuit **1402** responds to the select (SEL) signals from the Falling Edge Controller **1404** to couple bus **608** to a selected one or more device string **606** buses **906**. If the address data input on TDI does not match the address of the Falling Edge Controller **1404**, the ENA signal is not asserted and the TDO buffer and Routing Circuit are not enabled.

FIG. **18** illustrates one example implementation of the Falling Edge Controller (FEC) **1404** according to the present disclosure. However, the disclosure is not limited to only this one implementation example of FEC **1404**. In this one example, the FEC comprises a controller **1802**, a shift register **1804**, an update register **1806**, an address circuit **1808**, and TCK inverter **1810**, all connected as shown.

The controller **1802** has inputs for inputting the inverted TCK signal from bus **608**, the TMS signal from bus **608**, and the optional TRST input signal from bus **608**. The controller **1802** has outputs for outputting shift register control on bus **1812**, update register control on bus **1814**, the TRST output signal, and the Shift signal.

The shift register **1804** has inputs for inputting the TDI signal, the control signals **1812** from controller **1802**, the address and f (SEL) signals **1818** from the update register

1806 and has outputs for outputting address and select (SEL) signals **1816** to update register **1806** and for outputting data to TDO signal **1708**.

The update register **1806** has inputs for inputting the address and select signals **1816** from shift register **1804**, the control signals on bus **1814** from controller **1802**, the TRST output signal from controller **1802**, and has outputs for outputting the address and select (SEL) signals **1816** from shift register **1804**.

The address circuit **1808** has inputs for inputting the address signals from update register **1806** and has an output for outputting the ENA signal. The address circuit **1808** may have additional inputs for optionally inputting a desired unique externally supplied address, as shown in dotted line.

FIG. **19** illustrates one example implementation of the address circuit **1808** according to the present disclosure which comprises an address comparator **1902** and optionally an internally supplied address circuit **1904**. Preferably, the internally supplied address circuit **1904** is circuit that can be programmed or otherwise set to a desired unique address. The address comparator **1902** compares the address input from the update register **1806** to the address from the internally supplied address circuit **1904**. If the addresses match the ENA signal is asserted. If the addresses do not match the ENA signal is not asserted. As seen in FIG. **19**, the internally supplied address may be replaced with the externally supplied address if desired. If an externally supplied address is used the internally supplied address circuit **1904** can be removed from the address circuit **1808**.

FIG. **20** illustrates one example state diagram depicting the operation of controller **1802** comprising a Reset state, an Idle state, a Capture state, a Shift state, and an Update state. The controller **1802** transitions through these states in response the logic state of the TMS signal on the falling edge of the TCK signal. The controller **1802** will be transition to the Reset state if the optional TRST input signal is asserted or after a certain number of logic 1's have been input on the TMS signal.

In the Reset state, the TRST output from the controller is set low. The controller remains in the Reset state while TMS is high. In response to a low on the TRST output, the update register **1806** is reset to an address that does not match the address of the address circuit **1808**. Also in response to the low on the TRST output, TAPs **202** of devices **104** on a board are placed in the Test Logic Reset state of FIG. **4**. When TMS goes low, the controller transitions from the Reset state to the Idle state.

In the Idle state, the TRST output from the controller **1802** is set high to remove the reset condition from the update register **1806** and device TAPs **202**. The controller remains in the Idle state while TMS is low. While in the Idle state, the controller **1802** does not output shift register or update register control on buses **1812** and **1814**. The controller transitions from the Idle state to the Capture state when TMS goes high.

In the Capture state, the controller **1802** outputs control to the shift register **1804** on bus **1812** to cause the shift register to capture the address and select information output from the update register **1806** on bus **1818**. The controller transitions from the Capture state to the Shift state if TMS is low or transitions to the Reset state if TMS is high.

In the Shift state, the controller **1802** outputs control to the shift register on bus **1812** to cause the shift register to shift input data from the TDI input and shift output data to the TDO signal **1708**. The controller remains in the shift State while TMS is low and transitions to the Update state when TMS goes high. The data shifted in on TDI is the new

address and select information to be updated to update register **1806** and the data shifted out on TDO is the current address and select information contained in the update register **1806**. If the FER's Falling Edge Controller **1404** is not currently addressed or has been reset, the ENA output signal from the Address Circuit **1808** will not be asserted to enable the Routing Circuit **1502** or TDO output buffer **1702** signal. In this condition, the address and selection information on the TDO signal **1708** will not be output on the TDO signal of bus **608** since the output of TDO buffer **1702** is tri-state. If the FER's Falling Edge Controller **1404** is currently addressed the ENA signal will be asserted to enable TDO buffer **1702** to output the address and selection information from TDO signal **1708** to the TDO signal of bus **608**.

In the Update state, the controller **1802** outputs control to the update register **1806** on bus **1814** to cause the update register to load the address and select information on bus **1816** that was shifted into the shift register **1804** during the Shift state. From the Update state, the controller **1802** transitions back to the Idle state to wait for the next capture, shift and update operation.

As seen in the state diagram of FIG. **20**, the controller **1802** will transition to the Reset state from any other state if the TMS signal is set high for a number of falling edge TCK inputs.

FIG. **21** illustrates one example implementation of Routing Circuit **1502** according to the present disclosure. However, the disclosure is not limited to only this one implementation example of Routing Circuit **1502**. In this example, the Routing Circuit comprises; (1) a multiplexer **2102** for coupling TDI or TDOb to TDIA in response to a Select1 signal, (2) a gate **2104** for coupling TCK to TCKa in response to an Enable1 signal, (3) a gate **2106** for coupling TMS to TMSa in response to the Enable1 signal, (4) a multiplexer **2108** for coupling TDI or TDOa to TDIb in response to a Select2 signal, (5) a gate **2110** for coupling TCK to TCKb in response to an Enable2 signal, (6) a gate **2112** for coupling TMS to TMSb in response to the Enable 2 signal, (7) a multiplexer **2114** for coupling TDOa or TDOb to TDO **1710** in response to a Select3 signal, and a decode circuit **2116** having inputs for the SEL and ENA signals and outputs for the Enable1, Enable2, Select1, Select2, and Select3 signals. While this example shows gating both the TCKa and TMSa signals, gating of only the TCKa signal or only the TMSa signal may be used. Likewise gating of only the TCKb signal or only the TMSb signal may be used.

While the Routing Circuit **1502** of FIG. **21** is shown to couple bus **608** to a selected device string bus **906** using gating and multiplexing circuits, registration circuitry can be incorporated in the Routing Circuit **1502** to allow pipelining of the data (TDI and TDO) and control (TMS) signals between coupled buses.

FIG. **22** illustrates the operation of the decode circuit **2116** in response to the SEL and ENA signals from Falling Edge Controller **1404**.

If ENA is low the TCKa and/or TMSa signals are set low by the Enable1 signal and the TCKb and/or TMSb signals are set low by the Enable2 signal. In this condition access to the device strings **606** coupled to bus "a" **906** and bus "b" **906** is disabled.

If ENA is high and SEL is set to a first pattern (00) TDI is coupled to TDIA by Select1, TCK is coupled to TCKa by Enable1, TMS is coupled to TMSa by Enable1, TDOa is coupled to TDO **1710** by Select3, and TCKb and TMSb are set low by Enable2. In this condition access to the device

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string **606** coupled to bus “a” **906** is enabled and access to the device string **606** coupled to bus “b” **906** is disabled.

If ENA is high and SEL is set to a second pattern (01) TDI is coupled to TDIB by Select2, TCK is coupled to TCKb by Enable2, TMS is coupled to TMSb by Enable2, TDOb is coupled to TDO **1710** by Select3, and TCKa and TMSa are set low by Enable1. In this condition access to the device string **606** coupled to bus “b” **906** is enabled and access to the device string **606** coupled to bus “a” **906** is disabled.

If ENA is high and SEL is set to a third pattern (10) TDI is coupled to TDIA by Select1, TCK is coupled to TCKa and TCKb by Enable1 and Enable2, TMS is coupled to TMSa and TMSb by Enable1 and Enable2, TDOa is coupled to TDIB by Select2, and TDOb is coupled to TDO **1710** by Select3. In this condition access to both device strings **606** coupled to buses “a” and “b” **906** are enabled and placed in a serial arrangement, with device string **606** of bus “a” **906** being the first device string **606** in the serial arrangement.

If ENA is high and SEL is set to a fourth pattern (11) TDI is coupled to TDIB by Select2, TCK is coupled to TCKa and TCKb by Enable1 and Enable2, TMS is coupled to TMSa and TMSb by Enable1 and Enable2, TDOb is coupled to TDIA by Select1, and TDOa is coupled to TDO **1710** by Select3. In this condition access to both device strings **606** coupled to buses “a” and “b” **906** are enabled and placed in a serial arrangement, with device string **606** of bus “b” **906** being the first device string **606** in the serial arrangement.

FIG. **23** is provided to illustrate the operational states and timing of the FER **1402** and the device TAPs (DT) **202** in device string domains **606** of FIG. **14**. The operational states consist of; (1) a state **2302** where both the FER **1402** and DTs **202** are in a reset state (DTs in the Test Logic Reset state of FIG. **4** and the FER in the Reset state of FIG. **20**) in response to the TRST signal or logic values input on TMS, (2) a state **2304** where both the FER **1402** and the DTs **202** are in an idle state (DTs in the Run Test/Idle, Pause-DR or Pause-IR state of FIG. **4** and the FER in the Idle state of FIG. **20**) in response to logic values input on TMS, (3) a state **2306** where communication occurs to the FER **1402** while the DTs are idle in response to logic values input on TMS, and (4) a state **2308** where communication occurs to the DTs **202** while the FER is idle in response to logic values input on TMS.

Timing diagram **2310** illustrates that logic values input on TMS, indicated by darkened time slots, during the rising and falling edges of TCK in state **2304** maintain the FER **1402** and DTs **202** in idle state **2304**. In the idle state **2304**, no data input or data output occurs on TDI and TDO respectively, also indicated by darkened fill in timing diagram **2310**.

Timing diagram **2312** illustrates that logic values input on TMS (not darkened) during the falling edge of TCK in state **2306** enables the FER **1402** to input data from TDI and output data on TDO, while idle values on TMS (darkened) are input during the rising edge of TCK in state **2306** to maintain the DTs **202** in an idle state.

Timing diagram **2314** illustrates that logic values input on TMS (not darkened) during the rising edge of TCK in state **2308** enables the DTs **202** to input data from TDI and output data on TDO, while idle values on TMS (darkened) are input during the falling edge of TCK in state **2308** to maintain the FER **1402** in the idle state.

FIG. **24** illustrates an example system **2402** comprising multiple boards **602a-602b**. Each board contains a FER **1402** coupled to JTAG device strings **606**. The TDI input signal of the FERs of boards **602a** and **602b** are connected together and to a TDI output signal from a JTAG controller **106** via bus **608**. The TCK input signal of the FERs of boards

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602a and **602b** are connected together and to a TCK output signal from the JTAG controller **106** via bus **608**. The TMS input signal of the FERs of boards **602a** and **602b** are connected together and to a TMS output signal from the JTAG controller **106** via bus **608**. The TDO output signal of the FERs of boards **602a** and **602b** are connected together and to a TDO input signal to the JTAG controller **106** via bus **608**. While not shown the FERs of boards **602a** and **602b** may each include optional TRST input signals which are connected together and to a TRST output signal from the JTAG controller **106** via bus **608**. Each FER **1402** is addressable by a unique internally or externally supplied address, as describe in FIGS. **18-20**.

It should be understood that while the system **2402** of FIG. **24** is described as being a system containing multiple FER **1402** equipped boards **602a-602b**, the system **2402** could also be any type of higher level electrical system containing multiple FER **1402** equipped lower level electrical subsystems **602a-602b**, such as an IC containing multiple FER **1402** equipped embedded core circuits **602a-602b** or a core circuit containing multiple FER **1402** equipped further embedded core circuits **602a-602b**. This will be the case in other similar example figures of this disclosure.

When the JTAG controller **106** needs to access a JTAG device string **606** on a first one of the boards **602a-602b**, it performs a first falling edge scan operation, as described in regard to FIGS. **18-20**, to the FERs **1402** of boards **602a-602b**. The first falling edge scan operation loads address and device string selection information into the update registers **1806** of each FER **1402**. The FER that has an address that matches the address loaded into the update register **1806** becomes enabled to allow the JTAG controller **106** to access the selected JTAG device string **606** of the enabled FER **1402** using JTAG scan operations.

When the JTAG controller **106** needs to access a JTAG device string **606** on a second one of the boards **602a-602b**, it performs a second falling edge scan operation, as described in regard to FIGS. **18-20**, to the FERs **1402** of boards **602a-602b**. The second falling edge scan operation loads address and device string selection information into the update registers **1806** of each FER **1402**. The FER **1402** having an address that matches the address loaded into the update register **1806** becomes enabled to allow the JTAG controller **106** to access the selected JTAG device string **606** of the enabled FER **1402** using JTAG scan operations.

When the JTAG controller **106** needs to access a JTAG device string **606** on a third one of the boards **602a-602b**, it performs a third falling edge scan operation, as described in regard to FIGS. **18-20**, to the FERs **1402** of boards **602a-602b**. The third falling edge scan operation loads address and device string selection information into the update registers **1806** of each FER **1402**. The FER **1402** having an address that matches the address loaded into the update register **1806** becomes enabled to allow the JTAG controller **106** to access the selected JTAG device string **606** of the enabled FER **1402** using JTAG scan operations.

Assuming that prior to the first falling edge scan operation mentioned above none of the Routers **1402** were enabled, the first falling edge scan operation would input new address and selection information to the shift registers **1804** of all Routers **1402** from TDI but none of the Routers **1402** would be able to output their existing address and selection information from their shift register **1804** on TDO. This is because the ENA signal of the Falling Edge Controllers **1404** of each Router **1402** is not asserted to enable the Router's TDO output buffer **1702**, as described in regard to FIG. **17-20**.

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During the second falling edge scan operation mentioned above, all Routers **1402** will input new address and select information from TDI and the Router enabled by the first falling edge scan operation will output its existing address and selection information on TDO, since its ENA signal was asserted following the first falling edge scan operation.

During the third falling edge scan operation mentioned above, all Routers **1402** will input new address and select information from TDI and the Router enabled by the second falling edge scan operation will output its existing address and selection information on TDO, since its ENA signal was asserted following the second falling edge scan operation.

From the above it is seen that each falling edge scan operation, except for the first one, inputs new address and selection information to all Routers **1402** from the JTAG controller **106** while the currently enabled Router **1402** outputs its existing address and selection information to the JTAG controller **106**. Thus each time the JTAG controller performs a falling edge scan operation it can inspect the address and selection information it receives on TDO to verify that the address and selection information received was from a currently enabled Router **1402**.

In FIG. **24**, if the system **2402** contains a reasonable number of boards **602a-602b** the JTAG controller **106** can operate bus **608** to access the selected board at a reasonable communication bandwidth. However if a large number of boards **602a-602b** exists in system **2402**, excessive loading will occur on bus **608**, due to the large number of FERs **1402** connected to bus **608**. This excessive loading will reduce the communication bandwidth between the JTAG controller **106** and selected board to unacceptable levels. A solution to the bus **608** loading problem is described in FIGS. **25** and **26** below.

FIG. **25** illustrates an example system **2502** comprising multiple boards **602a-602c** arranged in separate board groups **2504-2506**. Each board group **2504-2506** has a separate JTAG bus **2508-2510** that is coupled to a device referred to as a Partitioning Falling Edge Router (PFER) **2512**. The PFER **2512** is interfaced to a JTAG controller **106** via JTAG bus **608**. The PFER **2512** can have any number of separate JTAG buses **2508-2510** interfaced to any number of separate board groups **2504-2506**. As can be seen, the PFER **2512** allows systems **2502** with large numbers of boards **602a-602c** to partition the boards into a separate board groups **2504-2506**, each group containing only a subset of the overall number of system boards. The JTAG controller **106** can access the JTAG bus **2508-2510** of any board group **2504-2506** by communication to the PFER **2512**. Thus the PFER **2512** solves the system bus **608** loading problem mentioned above by providing separate low load busses **2508-2510** to separate system board groups **2504-2506**.

It should be understood that while the system **2502** of FIG. **25** is described as being a system containing multiple FER **1402** equipped board groups **2504-2506** coupled to a PFER **2512**, the system **2502** could also be; (1) a board containing multiple FER **1402** equipped IC groups **2504-2506** coupled to a PFER **2512**, (2) an IC containing multiple FER **1402** equipped embedded core circuit groups **2504-2506** coupled to a PFER **2512**, or (3) a core circuit containing multiple FER **1402** equipped further core circuit groups **2504-2506** coupled to a PFER **2512**. This will be the case for other similar figures in this disclosure.

FIG. **26** illustrates an example implementation of PFER **2512**. The PFER **2512** is the same as the FER **1402** of FIG. **17** with the exceptions that; (1) multiplexer **1704** has been removed from the TDO output path from the Routing Circuit **1502** which allows the TDO output **1710** from the Routing

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Circuit to be directly input to TDO buffer **1702**, and (2) a multiplexer **2602** has been inserted in the TDI input path to the Routing Circuit **1502**. Multiplexer **2602** inputs the TDI input signal from bus **608**, the TDO signal **1708** from the shift register **1804** of Falling Edge Controller **1404**, the Shift signal from the Falling Edge Controller **1404**, and outputs a TDI signal to the Routing Circuit **1502**.

During falling edge scan operations, TDI data is shifted into the shift register **1804** of Falling Edge Controller **1404** and TDO data **1708** is shifted from the shift register **1804**. During shifting, multiplexer **2602** is controlled by the Shift signal to allow the TDO signal **1708** from the Falling Edge Controller's shift register to be input to the Routing Circuit **1502**. The TDO signal **1708** input to the Routing Circuit from multiplexer **2602** is output on the TDI data output of one or more JTAG buses **2508-2510**. Also during the falling edge scan operation, the TDO of bus **608** is driven by the TDO signal **1710** from a currently selected FER **1402** of one of the JTAG buses **2508-2510**.

Referring back to FIG. **25**, it can be seen that when the JTAG controller **106** performs a falling edge scan operation, TDI data from the JTAG controller passes through the shift register **1804** of the PFER's Falling Edge Controller **1404** to be input to the shift registers **1804** of the FERs **1402** of one or more board groups **2504-2506** via JTAG buses **2508-2510**. Also during the falling edge scan operation, TDO data from the shift register **1804** of the currently enabled FER **1402** of a board group **2504-2506** is input to the Routing Circuit **1502** of the PFER **2512** and output to the JTAG controller **106** via the TDO signal of bus **608**. The TDI data input to the PFER's shift register **1804** contains address information to enable the PFER, as described in FIGS. **17-20**, and select information to select one or more of the JTAG buses **2508-2510** as described in FIGS. **18-21**. The TDI data input to the shift register **1804** of the FERs **1402** contains address information to enable one of the FERs, as described in FIGS. **17-20**, and select information to select one or more of the JTAG device strings **606** coupled to the enabled FER as described in FIGS. **18-21**. Simply put, the PFER **2512** is a circuit that lies in series between the JTAG controller **106** and the FERs **1402** of the board groups **2504-2506** that responds to falling edge scan operations to access one or more of the board groups **2504-2506** via JTAG buses **2508-2510**. The Routing Circuit **1502** of the PFER can serially concatenate board group busses **2508-2510** together to allow board groups to be accessed simultaneously, as described in the Routing Circuit **1502** description of FIGS. **21** and **22**.

FIG. **27** is provided to illustrate the operational states and timing of the Routers (R) (i.e. PFER **2512** and FER **1402**) and the device TAPs (DT) **202** in device string domains **606** of FIG. **25**. The operational states consist of; (1) a state **2702** where both the Routers (**1402** and **2512**) and DTs **202** are in a reset state (DTs in the Test Logic Reset state of FIG. **4** and the Routers in the Reset state of FIG. **20**) in response to the TRST signal or logic values input on TMS, (2) a state **2704** where both the Routers and the DTs **202** are in an idle state (DTs in the Run Test/Idle, Pause-DR or Pause-IR state of FIG. **4** and the Routers in the Idle state of FIG. **20**) in response to logic values input on TMS, (3) a state **2706** where communication occurs to the Routers while the DTs are idle in response to logic values input on TMS, and (4) a state **2708** where communication occurs to the DTs **202** while the Routers are idle in response to logic values input on TMS.

Timing diagram **2710** illustrates that logic values input on TMS, indicated by darkened time slots, during the rising and

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falling edges of TCK in state 2704 maintain the Routers and DTs 202 in the idle state 2704. In the idle state 2704, no data input or data output occurs on TDI and TDO respectively, also indicated by darkened fill in timing diagram 2710.

Timing diagram 2712 illustrates that values input on TMS (not darkened) during the falling edge of TCK in state 2706 enables the PFER 2512 to input data from a JTAG controller 106 and pass the data on to the TDI inputs of the FERs 1402, while TDO data from the currently enabled FER 1402 is input to the TDO input of the PFER 2512 to be passed on to the TDO input of the JTAG controller 106. During this falling edge scan operation, idle values are input on TMS (darkened) during the rising edge of TCK to maintain the DTs 202 in an idle state.

Timing diagram 2714 illustrates that values input on TMS (not darkened) during the rising edge of TCK in state 2708 enables the selected string 606 of one or more DTs 202 to input TDI data from a JTAG controller 106 via the PFER 2512 and FER 1402 and to output TDO data to the JTAG controller via the FER 1402 and PFER 2512. During this rising edge DT 202 string scan operation, idle values are input on TMS (darkened) during the falling edge of TCK to maintain the Routers (PFER and FER) in an idle state.

FIG. 28 illustrates a system 2802 comprising FER 1402 equipped boards 602 in separate board groups 2804-2810. The FERs 1402 of board group 2804 are coupled to a first selectable JTAG bus 2812 of a first PFER 2816 and the FERs of board group 2806 are coupled to a second selectable JTAG bus 2814 of the first PFER 2816. The FERs 1402 of board group 2808 are coupled to a first selectable JTAG bus 2818 of a second PFER 2822 and the FERs of board group 2810 are coupled to a second selectable JTAG bus 2820 of the second PFER 2822. The JTAG bus of the first PFER 2816 is coupled to a first selectable JTAG bus 2824 of a third PFER 2828 and the JTAG bus of the second PFER 2822 is coupled to a second selectable JTAG bus 2826 of the third PFER 2828. The JTAG bus of the third PFER 2828 is coupled to a JTAG controller 106 via JTAG bus 608.

Access to a device string 606 of a board 602 in board group 2804 is achieved by the JTAG controller 106 performing a falling edge scan operation, as previously described, to shift address and selection information into the shift register 1804 of the third PFER 2828, the shift register 1804 of the first PFER 2816 and the shift registers 1804 of the FERs 1402 of board group 2804. The FER 1402 having an address that matches the address shifted into its shift register 1804 is enabled to allow the JTAG controller 106 to access the selected device string 606 of the selected board 602 in the selected board group 2804 via the third PFER 2828, the first PFER 2816, and the enabled FER 1402, using rising edge JTAG scan operations.

Access to a device string 606 of a board 602 in board group 2806 is achieved by the JTAG controller 106 performing a falling edge scan operation, as previously described, to shift address and selection information into the shift register 1804 of the third PFER 2828, the shift register 1804 of the first PFER 2816 and the shift registers 1804 of the FERs 1402 of board group 2806. The FER 1402 having an address that matches the address shifted into its shift register 1804 is enabled to allow the JTAG controller 106 to access the selected device string 606 of the selected board 602 in the selected board group 2806 via the third PFER 2828, the first PFER 2816 and the enabled FER 1402, using rising edge JTAG scan operations.

Access to a device string 606 of a board 602 in board group 2808 is achieved by the JTAG controller 106 performing a falling edge scan operation, as previously

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described, to shift address and selection information into the shift register 1804 of the third PFER 2828, the shift register 1804 of the second PFER 2822 and the shift registers 1804 of the FERs 1402 of board group 2808. The FER 1402 having an address that matches the address shifted into its shift register 1804 is enabled to allow the JTAG controller 106 to access the selected device string 606 of the selected board 602 in the selected board group 2808 via the third PFER 2828, the second PFER 2822, and the enabled FER 1402, using rising edge JTAG scan operations.

Access to a device string 606 of a board 602 in board group 2810 is achieved by the JTAG controller 106 performing a falling edge scan operation, as previously described, to shift address and selection information into the shift register 1804 of the third PFER 2828, the shift register 1804 of the second PFER 2822 and the shift registers 1804 of the FERs 1402 of board group 2810. The FER 1402 having an address that matches the address shifted into its shift register 1804 is enabled to allow the JTAG controller 106 to access the selected device string 606 of the selected board 602 in the selected board group 2810 via the third PFER 2828, the second PFER 2822, and the enabled FER 1402, using rising edge JTAG scan operations.

As seen in FIG. 28, there can be multiple arrangements 2830 of board groups 2804-2806 coupled to a PFER 2816 and board groups 2808-2810 coupled to a PFER 2822. The PFER of each arrangement 2830 can be uniquely addressed using falling edge scan operations to allow the JTAG controller 106 to access a selected device string 606 of a selected FER 1402 using rising edge JTAG scan operations as described above.

Also as seen in FIG. 28, PFERS can be arranged in a hierarchical fashion which extends the access of a JTAG controller 106 to device strings 606 of remotely positioned boards 602.

FIG. 29 illustrates a system 2902 comprising FER 1402 equipped boards in separate board groups 2904, 2906 and 2908. The FERs 1402 of board group 2904 are coupled directly to a JTAG controller 106 via bus 608. The FERs of board groups 2906 and 2908 are coupled to the JTAG controller 106 via a PFER 2512 as previously described. In this arrangement the JTAG controller 106 can perform a falling edge scan operation to directly access a board in board group 2904, or to access a board in one or both of board groups 2906-2908 via the PFER 2512. As seen, FERs 1402 and PFER 2512 can compatibly exist on the same bus 608 to a JTAG controller 106.

FIG. 30 illustrates a system 3002 comprising strings 3004 and 3006 of one or more devices 3008 coupled to a PFER 2512. The PFER is coupled to a JTAG controller 106 via bus 608. The devices 3008, as shown in FIG. 31, are designed to include both rising edge circuitry (REC) 3102 and falling edge circuitry (FEC) 3104 coupled to the devices TDI, TCK, TMS and TDO signal leads. The rising edge circuitry 3102 includes the JTAG TAP 202 and optionally other types of circuitry. During rising edge TCK scan operations, the rising edge circuitry 3102 inputs data from TDI and outputs data on TDO. During falling edge TCK scan operations, the falling edge circuitry 3104 inputs data from TDI and outputs data on TDO. Examples of devices 3008 that include both rising and falling edge circuitry are described in pending patent disclosures TI-66079, TI-66140 and TI-68392 all incorporated herein by reference.

It should be understood that the system 3002 of FIG. 30 could be; (1) a board containing multiple strings of rising and falling edge operated IC devices 3008 coupled to a PFER 2512, (2) an IC containing multiple rising and falling

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edge operated embedded core circuit devices **3008** coupled to a PFER **2512**, or (3) a core circuit containing multiple rising and falling edge operated further core circuit devices **3008** coupled to a PFER **2512**. This will be the case for other similar figures in this disclosure.

FIG. **32** illustrates a larger system **3202** comprising multiple systems **3002**, each system **3002** having a PFER **2512** coupled to a JTAG controller **106** via bus **608**. The JTAG controller **106** can access any one of the system **3002** device strings **3004-3006** via the system's **3002** PFER **2512**. The JTAG controller **106** can also access a series of concatenated device strings **3004-3006** of a selected system **3002** via the system's PFER **2512**.

FIG. **33** illustrates a further larger system **3302** comprising separate groups **3304-3308** of larger systems **3202**, each separate group **3304-3308** of larger systems **3202** being coupled to a PFER **2512** via separate busses **3310-3314**. The PFER **2512** of the further larger system **3302** is coupled to a JTAG controller **106** via bus **608**. The JTAG controller **106** can access any one or more of the larger systems **3202** in the separate groups **3304-3308** by communicating with the PFER **2512** of further larger system **3302** using falling edge scan operations.

FIG. **34** illustrates a device **3402**, which in this example is an IC or an embedded core within an IC, comprising individually selectable TAP **202** circuit domains **3404-3408**. TAP circuit domain **3404** is the JTAG boundary scan TAP **202** of the IC or core **3402**, TAP circuit domain **3406** is a TAP **202** of a first core circuit within the IC or core **3402**, and TAP circuit domain **3408** is a TAP of a second core circuit within the IC or core **3402**. Each TAP **3404-3408** is coupled to a selectable JTAG bus of a FER **1402** via a control bus (C) consisting of a TMS and a TCK signal, an input bus (I) consisting of a TDI signal, and an output bus (O) consisting of a TDO signal. The FER **1402** is coupled to a JTAG controller **106** via a JTAG bus **608**. In response to a falling edge scan operation, the FER **1402** can couple one of the TAP circuits **3404-3408** to the JTAG controller **106**, as previously described, so it can be accessed by the JTAG controller during a rising edge JTAG scan operation. The JTAG boundary scan TAP **3404** is accessed to perform JTAG test operations on the device **3402**. The Core TAP **3406** is accessed to perform test, debug, and/or emulation operations on the associated core of the device **3402**. Core TAP **3408** is accessed to perform test, debug and/or emulation operations on the associated core **3408** of the device **3402**. As previously described, the FER **1402** may serially concatenate multiple TAP circuit domains **3404-3408** together so that they can simultaneously perform an operation during a rising edge JTAG scan operation.

FIG. **35** illustrates a system **3502**, which could be a board, an IC, or a core circuit within an IC, comprising multiple FER **1402** equipped devices **3402**. The JTAG bus of each FER **1402** of each device **3402** are connected together (TDI to TDI, TMS to TMS, TCK to TCK, and TDO to TDO) and to a JTAG controller **106** via bus **608**. The JTAG controller **106** can access any one of the TAPs **3404-3406** in any one device **3402** via the device's FER. The JTAG controller **106** can also access any serial combination of TAPs **3404-3408** in a device **3402** via the device's FER.

FIG. **36** illustrates a system **3602**, which could be a board, an IC, or a core circuit within an IC, comprising separate groups **3604-3608** of devices **3402**, each separate group **3604-3608** of devices **3402** being coupled to a PFER **2512** via separate busses **3610-3614**. The PFER **2512** of the system **3602** is coupled to a JTAG controller **106** via bus **608**. The JTAG controller **106** can access any one or more

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of the devices **3402** in the separate groups **3604-3608** by communicating with the PFER **2512** of the system **3602** and the FER **1402** of the devices using falling edge scan operations as previously described.

FIG. **37** illustrates a device **3702**, which in this example is an IC or an embedded core within an IC, comprising individually selectable modified TAP circuit domains **3704-3708**. The modified TAP circuit domains **3704-3708**, as shown in FIG. **38**, are designed to include both rising edge circuitry (REC) **3802** and falling edge circuitry (FEC) **3804** coupled to the devices TDI, TCK, TMS and TDO signal leads. The rising edge circuitry **3802** includes the JTAG TAP **202** and optionally other types of circuitry. During rising edge TCK scan operations, the rising edge circuitry **3802** inputs data from TDI and outputs data on TDO. During falling edge TCK scan operations, the falling edge circuitry **3804** inputs data from TDI and outputs data on TDO. Examples of modified TAP circuit domains **3704-3708** that include both rising and falling edge circuitry are described in aforementioned pending patent disclosures TI-66079, TI-66140 and TI-68392.

Modified TAP circuit domain **3704** operates as a JTAG TAP **202** to perform JTAG boundary scan test operations in device **3702** during rising edge scan operations and operates as another circuit to perform other operations during falling edge scan operations. Modified TAP circuit domain **3706** operates as a JTAG TAP **202** to perform test, debug and/or emulation operations on the associated core during rising edge scan operations and operates as another circuit to perform other types of operations on the associated core during falling edge scan operations. Modified TAP circuit domain **3708** operates as a JTAG TAP **202** to perform test, debug and/or emulation operations on the associated core during rising edge scan operations and operates as another circuit to perform other types of operations on the associated core during falling edge scan operations.

Each modified TAP domain **3704-3708** is coupled to a separate JTAG bus of a PFER **2512** via a control bus (C) consisting of a TMS and a TCK signal, an input bus (I) consisting of a TDI signal, and an output bus (O) consisting of a TDO signal. The PFER **2512** is coupled to a JTAG controller **106** via a JTAG bus **608**. During falling edge scan operations, TDI data from the JTAG controller **106** is communicated through the shift register **1804** of the PFER **2512** and into the falling edge circuitry **3804** of a selected modified TAP domain **3704-3708**, while TDO data from the selected modified TAP domain is communicated through the Routing Circuit **1502** of the PFER **2512** to the TDO input of the JTAG controller **106**. During rising edge scan operations, TDI data from the JTAG controller **106** is communicated through the Routing Circuit **1502** of the PFER **2512** and into the rising edge circuitry **3802** of a selected modified TAP domain **3704-3708**, while TDO data from the selected modified TAP domain **3704-3708** is communicated through the Routing Circuit **1502** of the PFER **2512** to the TDO input of the JTAG controller **106**. As previously described, the Routing Circuit **1502** of the PFER **2512** can serially concatenate multiple modified TAP domains **3704-3708** together to allow the multiple modified TAP domains **3704-3708** to operate simultaneously during either a rising edge scan operation or a falling edge scan operation.

FIG. **39** illustrates a system **3902**, which could be a board, an IC, or a core circuit within an IC, comprising multiple PFER **2512** equipped devices **3702**. The JTAG bus of each PFER **2512** of each device **3702** are connected together (TDI to TDI, TMS to TMS, TCK to TCK, and TDO to TDO) and to a JTAG controller **106** via bus **608**. The JTAG

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controller 106 can access any one of the modified TAP domains 3704-3706 in any one device 3702 via the device's PFER. The JTAG controller 106 can also access any serial combination of modified TAP domains 3704-3708 in a device 3702 via the device's PFER.

FIG. 40 illustrates a system 4002, which could be a board, an IC, or a core circuit within an IC, comprising separate groups 4004-4008 of devices 3702, each separate group 4004-4008 of devices 3702 being coupled to a PFER 2512 via separate busses 4010-4014. The PFER 2512 of the system 4002 is coupled to a JTAG controller 106 via bus 608. The JTAG controller 106 can access any one or more of the devices 3702 in the separate groups 4004-4008 by communicating with the PFER 2512 of system and the PFER 2512 of the devices 3702 using rising and falling edge scan operations as previously described.

FIG. 41 illustrates an example implementation of a Configurable Falling Edge Router (CFER) 4102 that is programmable, via a Mode signal, to operate as either the FER 1402 or the PFER 2512. The CFER 4102 has a modified Falling Edge Controller 1404, a Routing Circuit 1502, multiplexer 2602, multiplexer 1704, And gate 4108, And gate 4112 and TDO output buffer 1702. The Falling Edge Controller 1404 of FIG. 18 is modified in FIG. 41 to include a Mode output signal 4110. The Mode output signal 4110 controls whether the CFER 4102 operates as a FER 1404 or a PFER 2512.

FIG. 42 illustrates an example modification of the Falling Edge Controller 1404. The modification is simply to extend the shift register 1804 of the Falling Edge Controller 1404 to include a bit position 4202 for inputting the Mode signal 4110 and to extend the update register 1806 of the Falling Edge Controller to include a bit position 4204 for outputting the Mode signal 4110.

When the Mode signal 4110 is set low, the Shift signal from the Falling Edge Controller 1404 passes through And gate 4108 during falling edge scan operations (i.e. during the Shift state of FIG. 20) to allow the data TDO signal 1708 from Shift register 1804 to be output on TDO of bus 608, via multiplexer 1704. Also when the Mode signal 4110 is set low, And gate 4112 forces multiplexer 2602 to pass the TDI signal of bus 608 to the Routing Circuit 1502. In this configuration, the CFER 4102 operates identical to the FER 1402 of FIG. 17.

When the Mode signal 4110 is set high, And gate 4108 forces the multiplexer 1704 to output the TDO data 1710 from the Routing Circuit 1502 to the TDO of bus 608 during falling edge scan operations. Also when the Mode signal 4110 is set high, the Shift signal from the Falling Edge Controller 1404 passes through And gate 4112 during falling edge scan operations (i.e. during the Shift state of FIG. 20) to allow the data TDO signal 1708 from Shift register 1804 to be input to the Routing Circuit 1502, via multiplexer 2602. In this configuration, the CFER 4102 operates identical to the PFER 2512 of FIG. 26.

As seen in dotted line, And gate 4112 may be removed from the CFER 4102 to allow the Shift signal from the Falling Edge Controller 1404 to be directly connected to the control input of multiplexer 2602. While this changes the operation of the CFER 4102 from being identical to the FER 1402, since the data TDO signal 1708 from Shift register 1804 during the Shift state of FIG. 20 is input to the Routing Circuit instead of the TDO data of bus 608, it does not affect the ability of the CFER 4102 to operate as a FER 1402.

FIG. 43 illustrates a system 4301 with a CFER 4102 operating as a FER 1402 to interface a JTAG controller 106 to a first device or device string 4302 and a second device or device string 4304. The Mode signal 4110 of the CFER

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4102 is set low to enable this operation mode. Preferably, but not necessarily, according to the disclosure, the Mode signal 4110 is set low in response to a TRST signal input to the update register 1806, to allow the CFER 4102 to be immediately configured as a FER 1402 in response to the TRST signal. The TRST signal input to the update register 1806 may occur in response to an external TRST input of FIG. 18, entry into the Reset state of FIG. 20, or by a power up reset circuit associated with CFER 4102. The advantage of having the Mode signal 4110 initially set low is that the CFER 4102 is immediately available for use as a FER 1402 to allow a JTAG controller 106 to access a device or device string 4302-4304. The system 4301 may be a board, an IC or an embedded core circuit within an IC.

FIG. 44 illustrates a further system 4401 with a CFER 4402 interfacing a JTAG controller 106, to multiple systems 4301 of FIG. 43. The Mode signal of CFER 4402 is set high to enable it to operate as a PFER 2512, and the Mode signals of the CFERS in systems 4301 are set low to enable them to operate as FERs 1402. If the Mode signal 4110 of CFER 4402 is initially set low after a TRST signal input as described in FIG. 43, an initial falling edge scan operation to the CFER 4402 from the JTAG controller 106 will be necessary to configure CFER 4402 to operate as a PFER 2512. The further system 4401 may be a board, an IC or an embedded core circuit within an IC.

As seen in FIGS. 43 and 44, the advantage of the CFER 4102 over a separate FER 1402 and PFER 2512 is that only one product, the CFER 4102, need be provided by a semiconductor manufacturer to support the functionality of both the FER 1402 and PFER 2512. Customers purchasing the CFER 4102 can selectively use the CFER 4102 in their systems as either a FER 1402 or PFER 2512.

FIG. 45 illustrates a JTAG controller 106 TDI connection to shift registers 1804 in FERs 4502-4504, which can be FERs 1402 or CFERS 4102. The purpose of this illustration is to show the advantage of using equal length shift registers 1804 in FERs 4502-4504. Assuming the shift registers 1804 are designed to have the same fixed number of bits, the JTAG controller 106 simply inputs the fixed number of bits to the shift registers 1804 via TDI during the Shift state of FIG. 20 to enable one of the FERs 4502-4504 and select a device 104 or device string 606 coupled to the enabled FER 4502-4504. If the shift registers 1804 had different bit lengths the JTAG controller would have to shift in a different number of bits each time a different FER is to be enabled for accessing a device 104 or device string 606, which complicates the JTAG controller software. Also if the shift registers 1804 had different bit lengths it opens up the possibility that two or more FERs 4502-4504 may accidentally be addressed and enabled together. This would be due to a bit pattern shifted into a shift register 1804 of a desired FER 4502-4504 to be addressed and enabled having a subset bit pattern that also addresses and enables a non-desired FER 4502-4504.

FIG. 46 illustrates a JTAG controller 106 TDI connection to shift registers 1804 in PFERs 4602-4604, which can be PFERs 2512 or CFERS 4102. The PFERs 4602-4604 each pass the TDI data from the JTAG controller onto a group of connected FERs 4502-4504. The purpose of this illustration is to show the advantage of using equal length shift registers 1804 in both the PFERs 4602-4604 and FERs 4502-4504. Assuming the shift registers 1804 of both the PFERs 4602-4604 and FERs 4502-4504 are designed to have the same fixed number of bits, the JTAG controller 106 simply performs a scan operation containing the fixed number of bits for the shift registers 1804 in the PFERs and the fixed number of bits for the shift registers 1804 in the connected

FERs to address and enable a selected PFER **4602-4604** and one or more of its connected FERs **4502-4504**. If the shift registers **1804** of the PFERs **4602-4604** and FERs **4502-4504** had different bit lengths the JTAG controller would have to shift in a different number of bits each time a different PFER and FER combination is to be enabled for accessing a device **104** or device string **606**, which complicates the JTAG controller software. Also if the shift registers **1804** of the PFER and FER had different bit lengths it opens up the possibility that two or more PFER and FER combinations may accidentally be addressed and enabled together. This would be due to a bit pattern shifted into the shift registers **1804** of a desired PFER and FER combination to be addressed and enabled having a subset bit pattern that also addresses and enables a non-desired PFER and FER combination.

According to the disclosure, the shift registers **1804** of both the PFERs **4602-4604** and FERs **4502-4504** should preferably be fixed at the same length. However, the fixed shift register length of the PFERs **4602-4604** may be different from the fixed shift register length of the FERs **4502-4504** if desired without incurring the above mentioned problem.

While this disclosure has described use of a FER **1402**, PFER **2512**, or CFER **4102** that operates in response to the falling edge of TCK to access a rising edge TCK operated JTAG port on a device or device string, the disclosure is not limited to accessing only JTAG ports. In general, the router of this disclosure can be used to access any type of clocked device port by using the inactive edge of the ports clock, as described in following FIGS. **47** and **48**.

FIG. **47** illustrates a system **4701** comprising an Inactive Edge Router **4706** and Active Edge Ports **4702-4704**. The Active Edge Ports **4702-4704** may be any type of input/output port that exists within the system **4701**. The system **4701** may be a board, an IC, or an embedded core circuit within an IC. A Port Controller **4708**, which may be any type of port controller, is coupled to the Inactive Edge Router **4706** via a bus **4710** comprising a data input (DI) signal in place of the TDI signal, a clock (CK) signal in place of the TCK signal, a mode select (MS) signal in place of the TMS signal, and a data output (DO) signal in place of the TDO signal. The Active Edge Ports **4702** and **4704** are coupled to the Inactive Edge Router **4706** via buses **4712** and **4714** respectively. Each bus **4712** and **4714** comprises a DI signal, a CK signal, a MS signal, and a DO signal. The words "Active Edge" indicates the CK edge the port uses to perform its input/output operation. The words "Inactive Edge" indicates the CK edge the Router **4706** uses to couple Active Edge Port **4702** and/or Active Edge Port **4704**, via buses **4712** and **4714**, to the Port Controller **4708** via bus **4710**. If the "Active Edge" is the rising edge of the CK signal the "Inactive Edge" is the falling edge of the CK signal. If the "Active Edge" is the falling edge of the CK signal the "Inactive Edge" is the rising edge of the CK signal. In this example, the operation of the Inactive Edge Router **4706** is assumed to be the same as the Falling Edge Router **1402** described in regard to FIGS. **17-23**. FIG. **49A** illustrates one example implementation of the Inactive Edge Router **4706** which is based on the FER **1402** architecture of FIG. **17**. Using the Inactive Edge Router **4706** the Port Controller **4708** can perform an "Inactive Edge" scan operation, as previously described using "Falling Edge" scan operations, to select one or more of the Active Edge Ports to be accessed using "Active Edge" scan operations.

FIG. **48** illustrates a further system **4802** comprising a Partitioning Inactive Edge Router **4804** and first and second

systems **4701**. The further system **4802** may be a board, an IC, or an embedded core circuit within an IC. The Port Controller **4708** is coupled to the Partitioning Inactive Edge Router **4804** via a bus **4710** comprising a DI signal, a CK signal, a MS signal, and a DO signal. The systems **4701** are coupled to the Partitioning Inactive Edge Router **4804** via buses **4806** and **4808** respectively. Each bus **4806** and **4808** comprises a DI signal, a CK signal, a MS signal, and a DO signal. The definitions of wordings "Active Edge" and "Inactive Edge" are the same as mentioned in FIG. **47**. In this example, the operation of the Partitioning Inactive Edge Router **4804** is assumed to be the same as the Partitioning Falling Edge Router **2512** described in regard to FIGS. **26** and **27**. FIG. **49B** illustrates one example implementation of the Partitioning Inactive Edge Router **4804** which is based on the PFER **2502** architecture of FIG. **26**. Using the Partitioning Inactive Edge Router **4804** the Port Controller **4708** can perform an "Inactive Edge" scan operation, as previously described using "Falling Edge" scan operations, to select one or more of the Active Edge Ports **4702-4704** in one or more of the systems **4701** to be accessed using "Active Edge" scan operations.

It should be understood that a Configurable Inactive Edge Router could be designed and used in place of the Inactive Edge Router **4706** and Partitioning Inactive Edge Router **4804** of FIGS. **47** and **48**, as the Configurable Falling Edge Router **4102** was described replacing the Falling Edge Router **1402** and Partitioning Falling Edge Router **2512** in FIGS. **41-44**. FIG. **49C** illustrates one example implementation of a Configurable Inactive Edge Router **4902** which is based on the CFER **4102** architecture of FIG. **41**.

The Inactive Edge Controllers **1404** in the Inactive Edge Routers of FIGS. **49A-49C** will be designed to operate on the opposite clock edge that operates the Active Edge Ports **4702-4704**. This can be achieved by simply including or excluding inverter **1810** of FIG. **18** on the CK input to the Inactive Edge Controllers **1404**.

Although the disclosure has been described in detail, it should be understood that various changes, substitutions and alterations may be made without departing from the spirit and scope of the disclosure as defined by the appended claims.

What is claimed is:

1. A device comprising:
 - a first test access port (TAP);
 - a second TAP;
 - a test clock (TCK) terminal configurable to receive a TCK signal;
 - a test mode select (TMS) terminal; and
 - a router circuit coupled to the first TAP, the second TAP, the TCK terminal, and the TMS terminal, wherein the router circuit includes an inverter circuit coupled to the TCK terminal and configurable to invert the TCK signal, and wherein the router circuit is configurable to: receive a TCK signal via the TCK terminal, wherein the TCK signal received at the TCK terminal includes a series of alternating active edges and inactive edges; and select the second TAP based on communication received from the TMS terminal at an inactive edge of the TCK signal received at the TCK terminal.
2. The device of claim 1, wherein the router circuit is configurable to:
 - receive a TMS signal via the TMS terminal; and
 - connect the second TAP to the TMS terminal in response to selecting the second TAP.

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3. The device of claim 2, wherein the router circuit is configurable to connect the first TAP to the TMS terminal before selecting the second TAP.

4. The device of claim 1, wherein the first TAP is configurable to:

receive a TMS signal via the TMS terminal; and
perform a boundary scan test based on the TMS signal.

5. The device of claim 4, wherein the first TAP is configurable to perform the boundary scan test based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

6. The device of claim 1, wherein the second TAP is configurable to:

receive a TMS signal via the TMS terminal; and
perform a debug operation based on the TMS signal.

7. The device of claim 6, wherein the second TAP is configurable to perform the debug operation based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

8. The device of claim 1, wherein the second TAP is configurable to perform an emulation operation based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

9. The device of claim 1, wherein the second TAP is configurable to perform a programming operation based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

10. The device of claim 1, further comprising a third TAP coupled to the router circuit,

wherein the router circuit is configurable to select the third TAP based on communication received from the TMS terminal at an inactive edge of the TCK signal received at the TCK terminal.

11. The device of claim 10, wherein the third TAP is configurable to:

receive a TMS signal via the TMS terminal; and
perform a debug operation based on the TMS signal.

12. The device of claim 11, wherein the third TAP is configurable to perform the debug operation based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

13. The device of claim 10, wherein the third TAP is configurable to perform an emulation operation based on communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

14. The device of claim 10, wherein the third TAP is configurable to perform a programming operation based on

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communication received from the TMS terminal at an active edge of the TCK signal received at the TCK terminal.

15. The device of claim 1,

wherein the active edges of the TCK signal are rising edges, and

wherein the inactive edges of the TCK signal are falling edges.

16. The device of claim 1, wherein on power-up of the device, the router circuit is configurable to select the first TAP and disable the second TAP.

17. The device of claim 1, wherein the first TAP comprises a Joint Test Action Group (JTAG) TAP in accordance with IEEE 1149.1.

18. The device of claim 1, wherein the second TAP comprises a core TAP.

19. A device comprising:

a first test access port (TAP);

a second TAP;

a test data in (TDI) terminal;

a test data out (TDO) terminal;

a test clock (TCK) terminal;

a test mode select (TMS) terminal; and

a router circuit coupled to the first TAP, the second TAP, the TDO terminal, the TCK terminal, and the TMS terminal,

wherein the router circuit includes an inverter circuit coupled to the TCK terminal,

wherein the router circuit is configurable to:

receive a TCK signal via the TCK terminal, wherein the TCK signal received at the TCK terminal includes a series of alternating rising edges and falling edges; and

select the second TAP in response to communication received from the TMS terminal at a falling edge of the TCK signal received at the TCK terminal, and

wherein the first TAP is configurable to:

input data from the TDI terminal at a first rising edge of the TCK signal received at the TCK terminal; and
output data to the TDO terminal at a second rising edge of the TCK signal received at the TCK terminal.

20. The device of claim 19, wherein the router circuit is configurable to:

connect the first TAP to the TMS terminal before selecting the second TAP; and

connect the second TAP to the TMS terminal in response to selecting the second TAP.

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